GWASH vs. LDSC Intercept Choice

GWASH, LDSC free intercept, and LDSC fixed intercept compared together, under real chr1/chr22 LD structure

Executive Summary

This report compares GWASH's heritability estimator directly against the two standard LDSC intercept choices (free intercept and fixed intercept, both weighted and using the two-step procedure - the default LDSC settings used as the main estimators throughout this project's companion reports). The comparison spans every scale (n=500/1000/2000/5000), chromosome (chr1, chr22), and the project's four core scenarios (reference panel present or not, stratification vector f shared or drawn independently).

Headline finding: in the three well-specified scenarios (no reference panel with either f-sharing choice, and a reference panel with *shared* f), GWASH and both LDSC intercept choices track each other and the theory line closely across the full F_{ST} range. In the pathological scenario - a reference panel whose stratification vector f is mismatched with the main sample - the two estimators respond completely differently as sample size grows. At F_{ST}=0.5, chr1, LDSC's free-intercept estimate rises from 0.87 (n=500) to 4.53 (n=5000), and the fixed-intercept estimate rises from 2.56 to 6.04 - both scaling up sharply with n. GWASH, over that same range of scales, stays essentially flat at 0.51-0.59 (roughly 1.7-2× h2_pop, not 15-20×). GWASH is still biased upward by the mismatch, but unlike LDSC, that bias does not appear to grow with sample size.

Methods

See the companion reports (realistic_LD_sharedf_vs_indepf_summary.pdf and ldsc_estimator_variants_comparison.pdf) for full simulation methodology. The three estimators compared here, all computed from the same per-replicate data (same X, y, LD scores):

- **h2_gwash**: GWASH
- **h2_ldsc_reg**: LDSC, free intercept (weighted, two-step)
- **h2_ldsc_fixed**: LDSC, fixed intercept (weighted, two-step)

All panels use a linear y-axis. The grey dashed line is h2_pop=0.3, the true simulated genetic heritability. The black dotted line is the theoretical total h2 (h2_pop + sigma_s^2/C_f, C_f = mean(f^2/(1+f^2))), using that scenario's own per-replicate f draws.

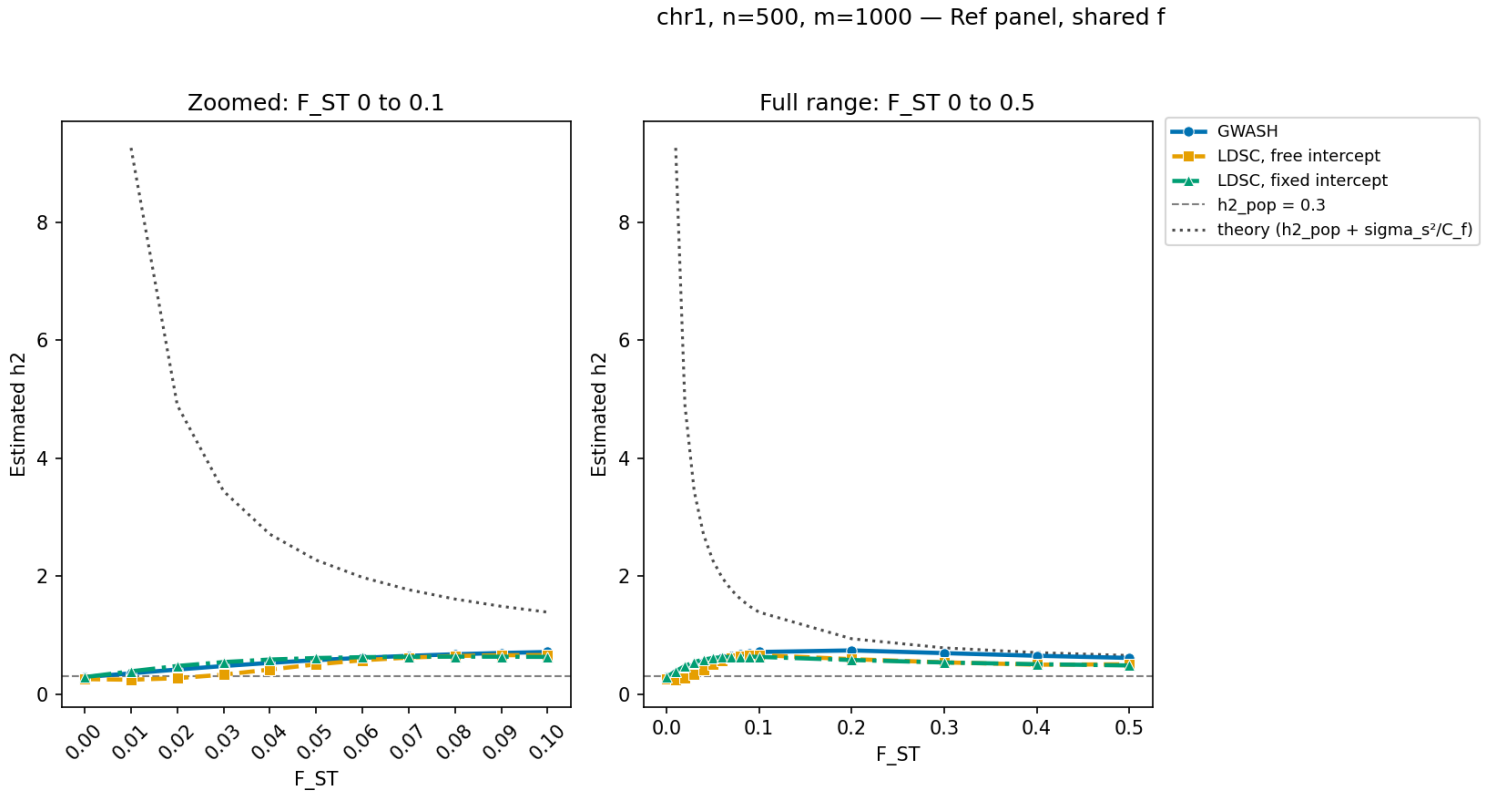
Summary: GWASH vs. LDSC at F_ST=0.5, Reference Panel + Independent f

Mean h2 estimate (100 replicates) for each estimator at F_{ST}=0.5, in the scenario where the companion report found the largest divergence: a reference panel whose f does not match the main sample's.

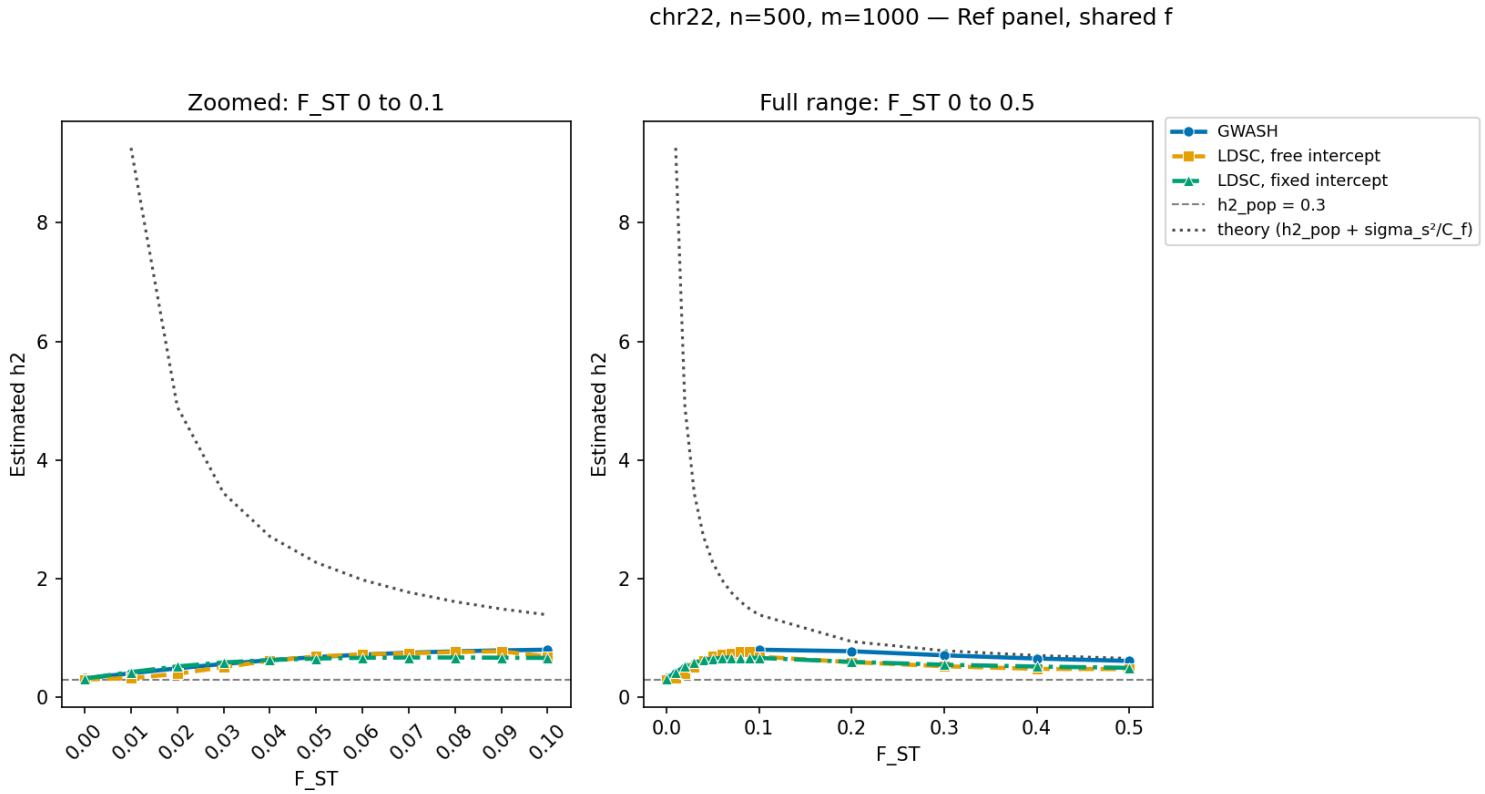
Chrom	n	GWASH	LDSC free intercept	LDSC fixed intercept
chr1	500	0.526	0.869	2.560
chr1	1000	0.521	1.512	3.205
chr1	2000	0.509	2.532	4.301
chr1	5000	0.588	4.529	6.037
chr22	500	0.518	0.967	3.005
chr22	1000	0.514	1.478	3.314
chr22	2000	0.514	2.481	4.315
chr22	5000	0.584	4.394	5.865

Both LDSC intercept choices inflate sharply with scale in this scenario (consistent with the two-step finding in the LDSC-variants companion report). GWASH, by contrast, stays close to a roughly constant 0.5-0.6 across all four scales on both chromosomes - biased above h2_pop=0.3, but not scale-dependent the way LDSC is.

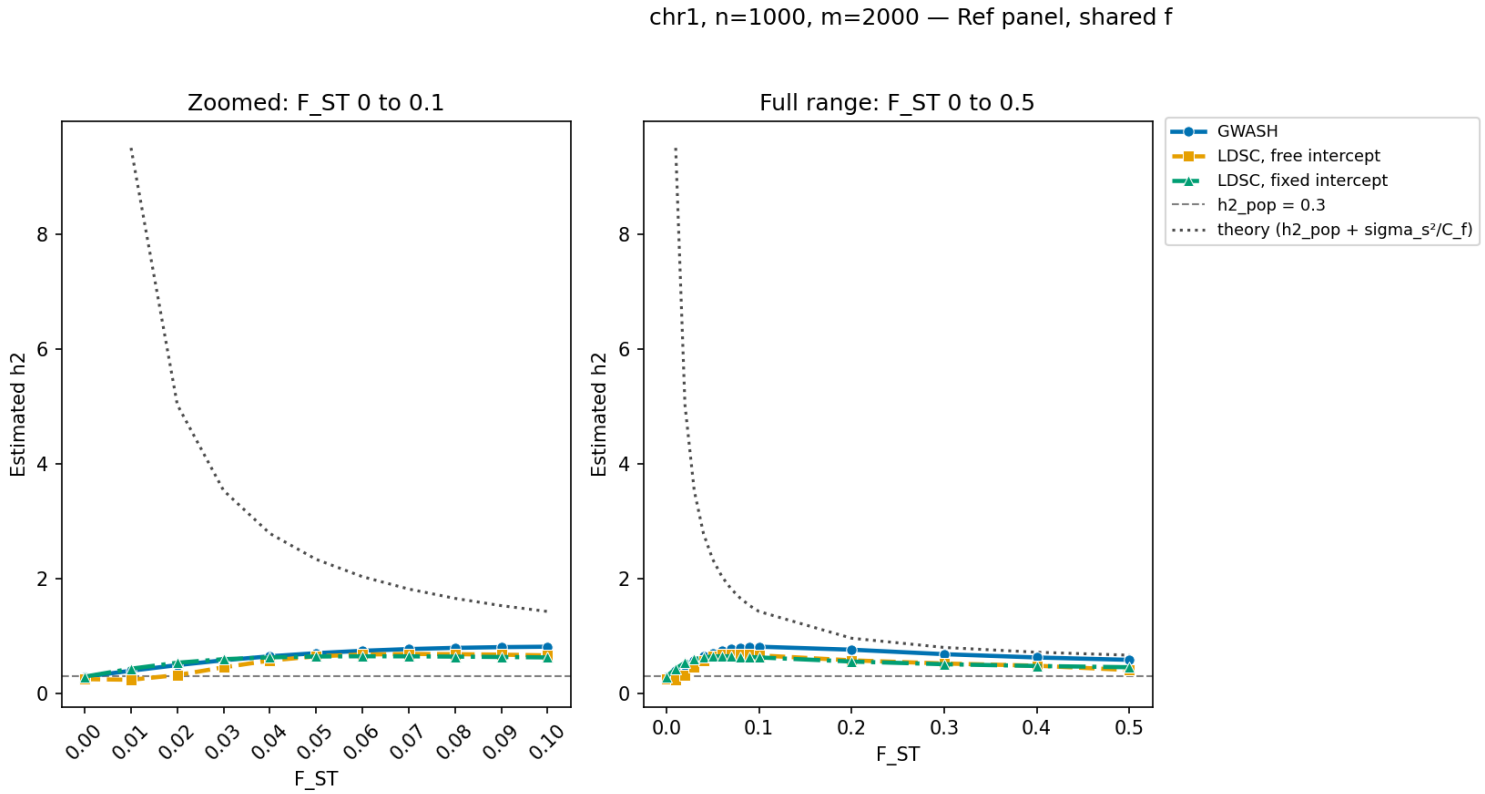
Figures: Ref panel, shared f



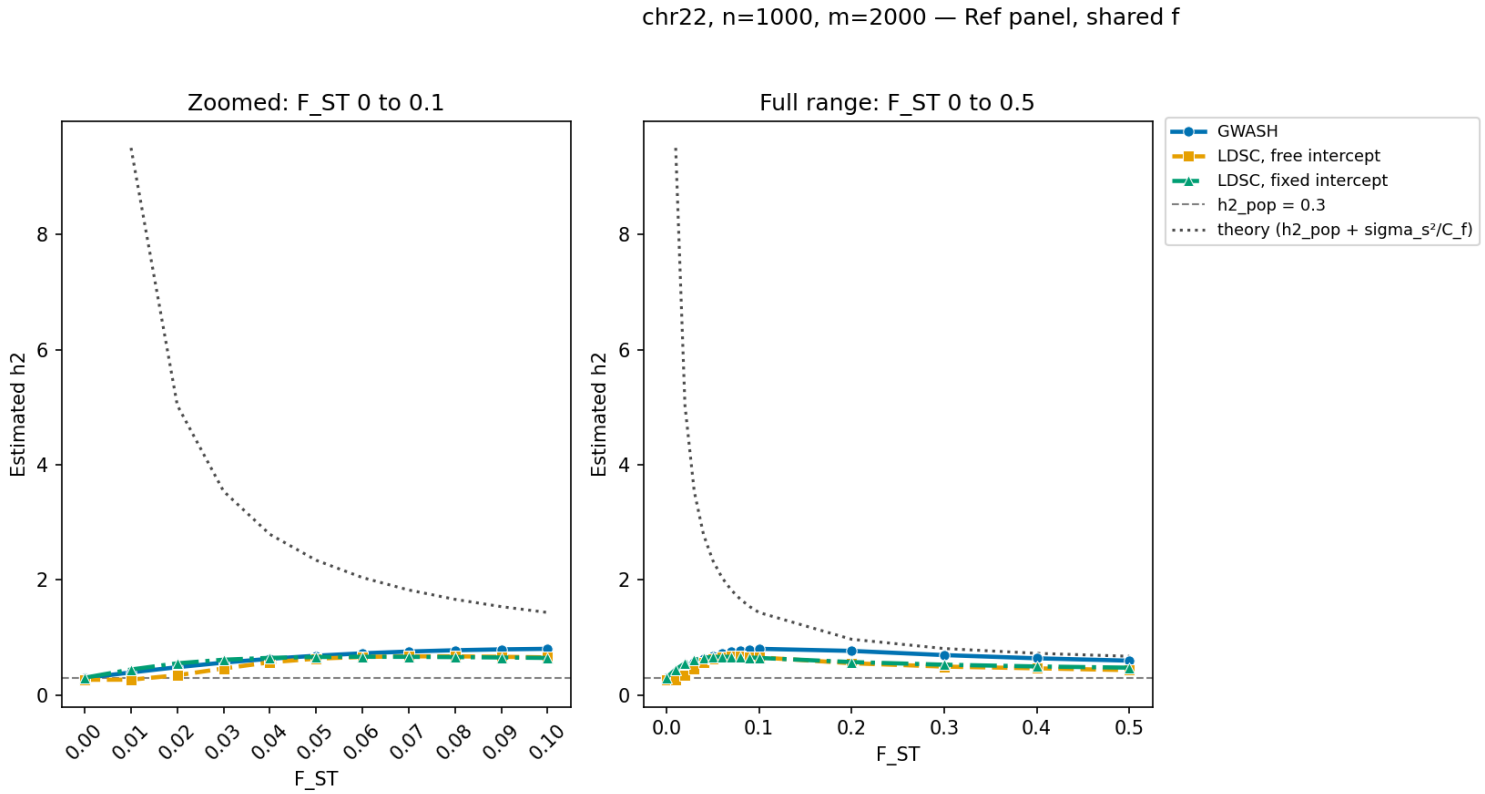
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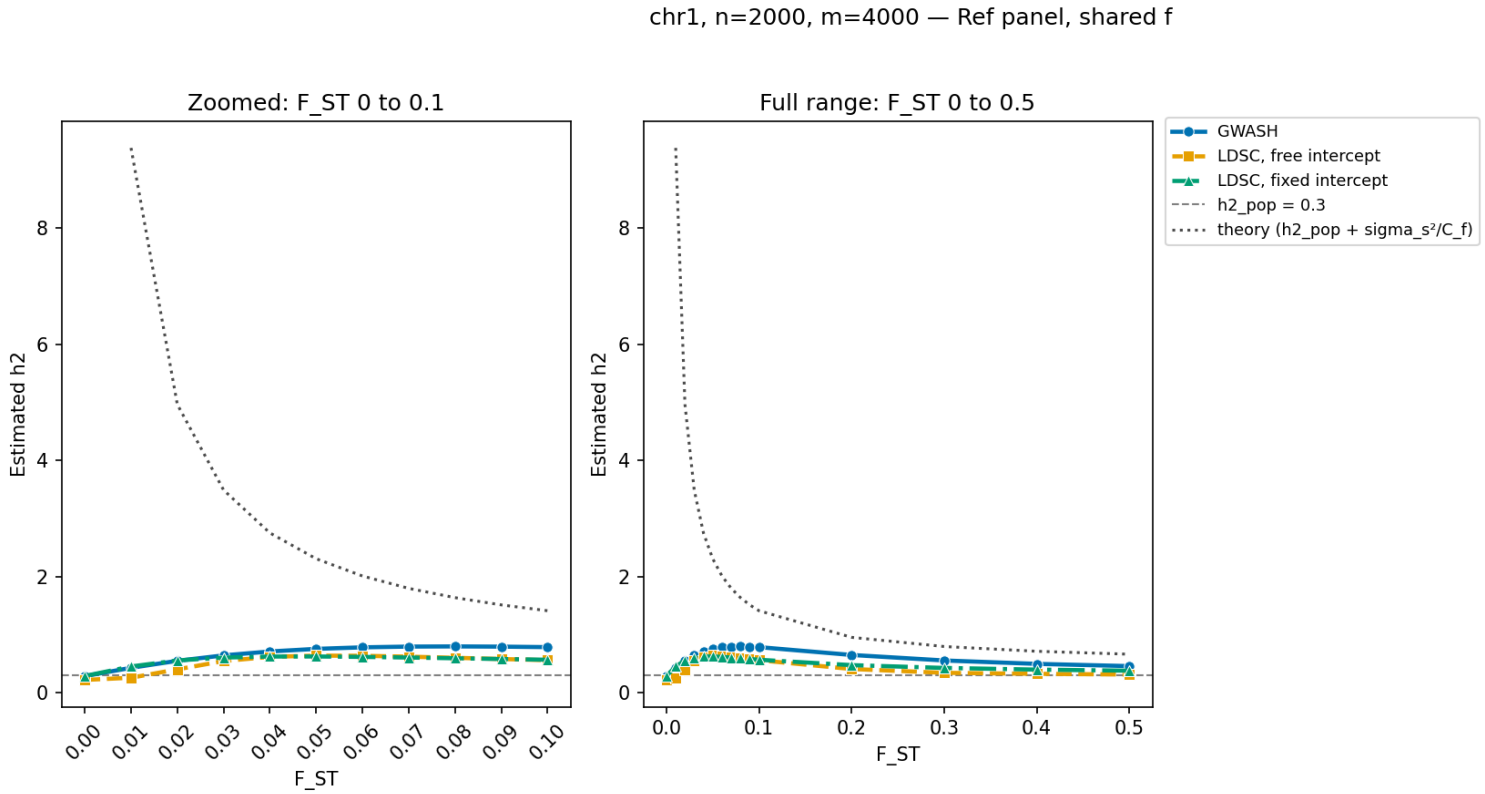
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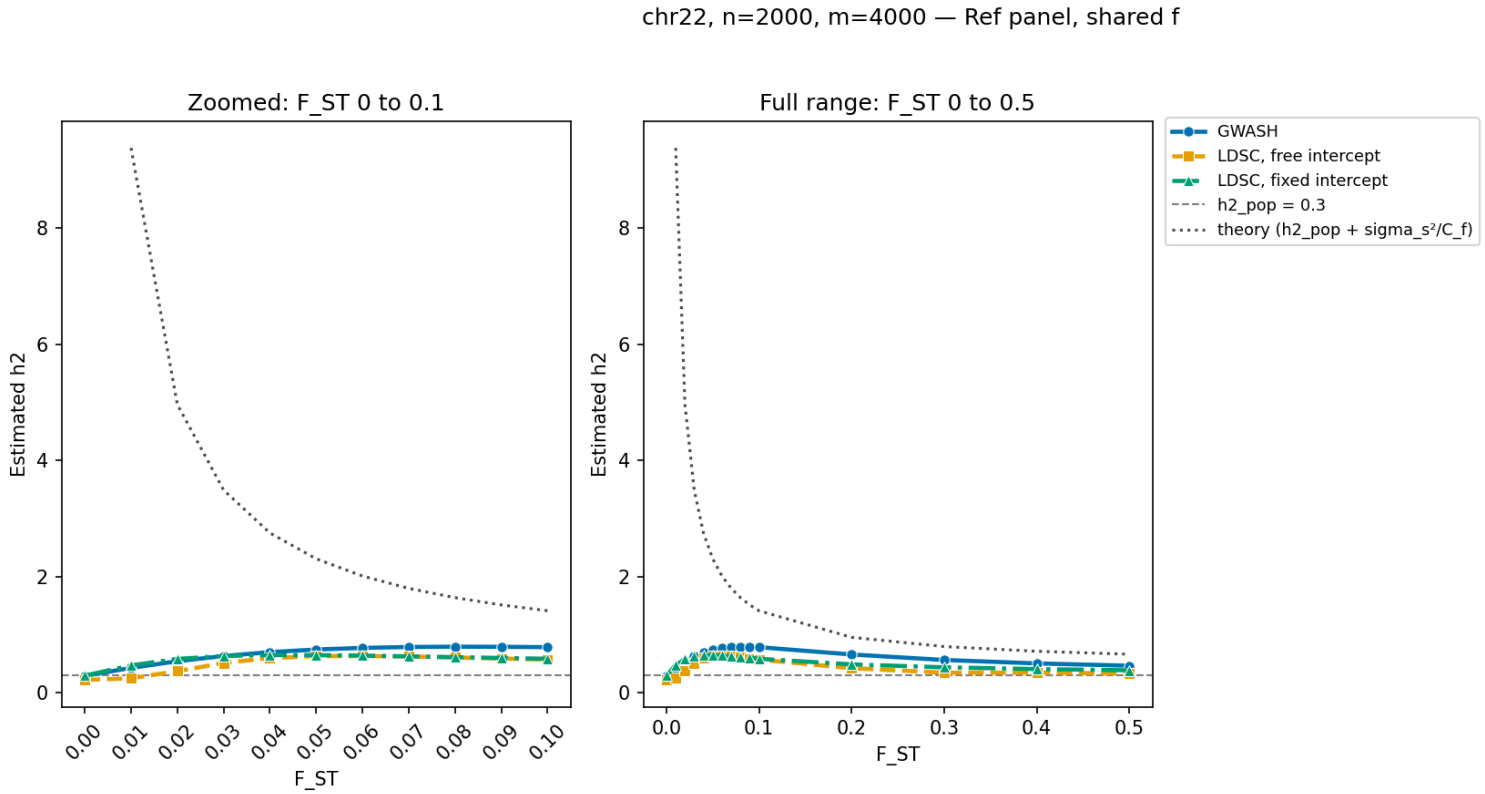
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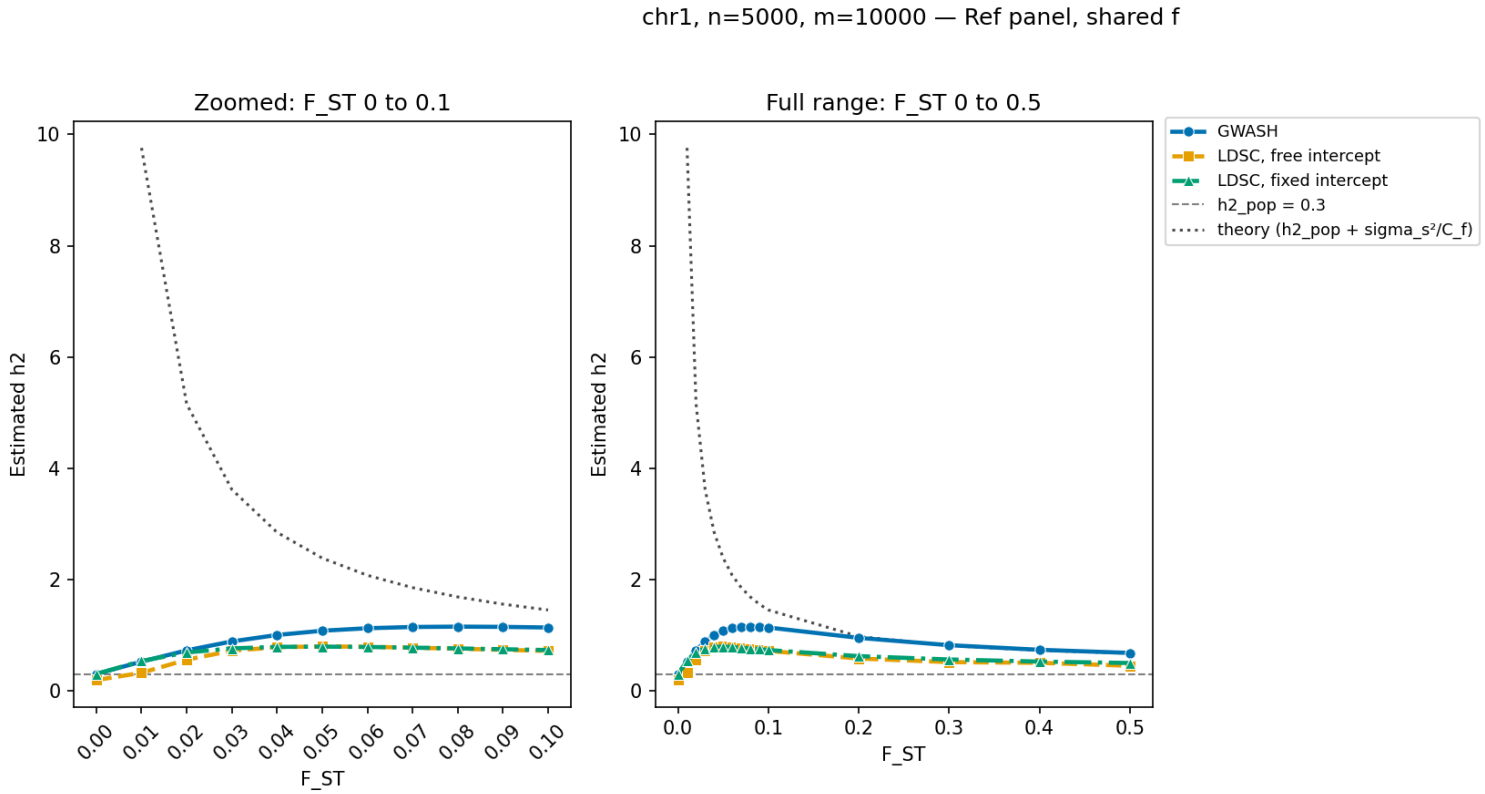
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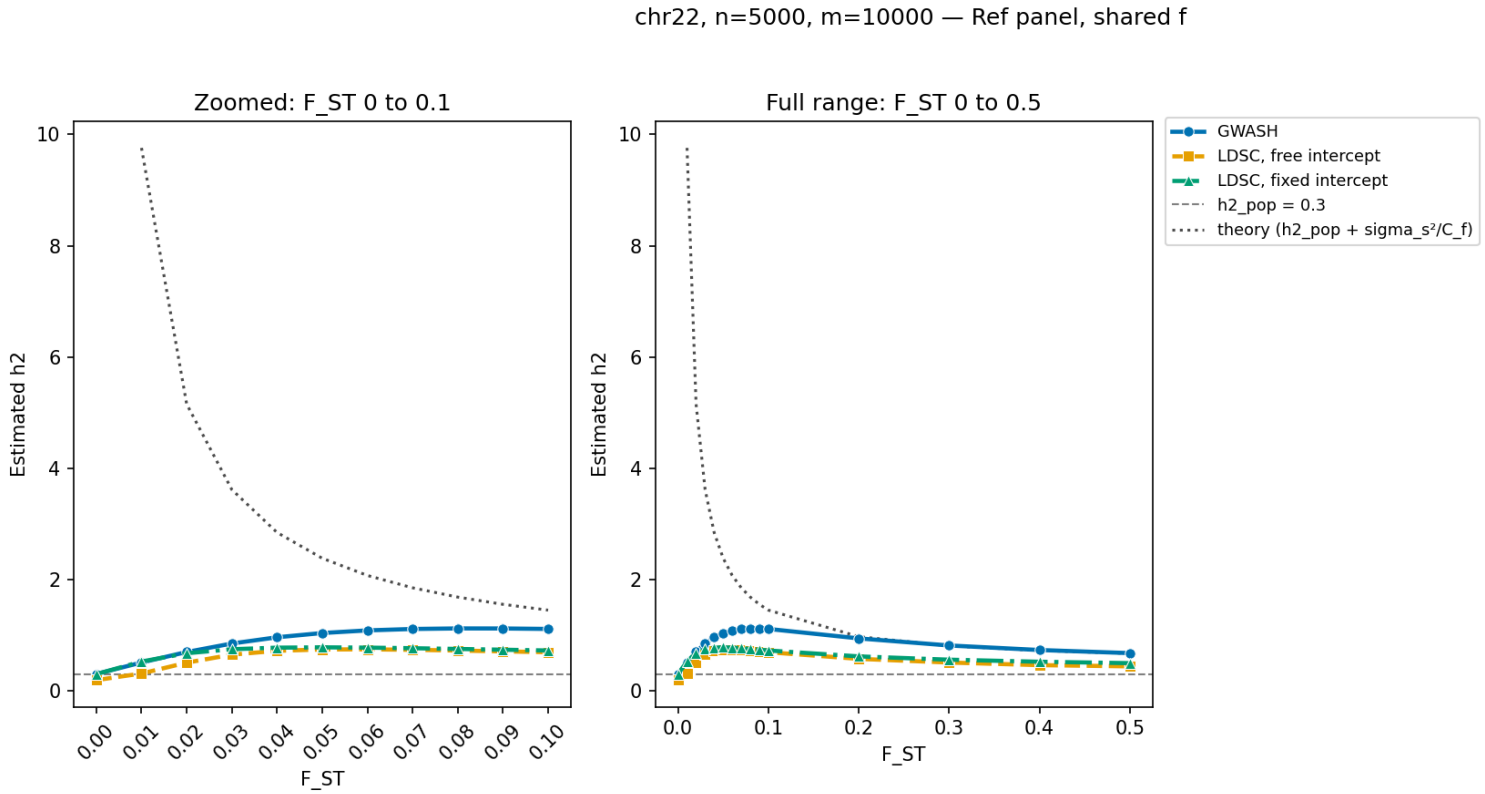
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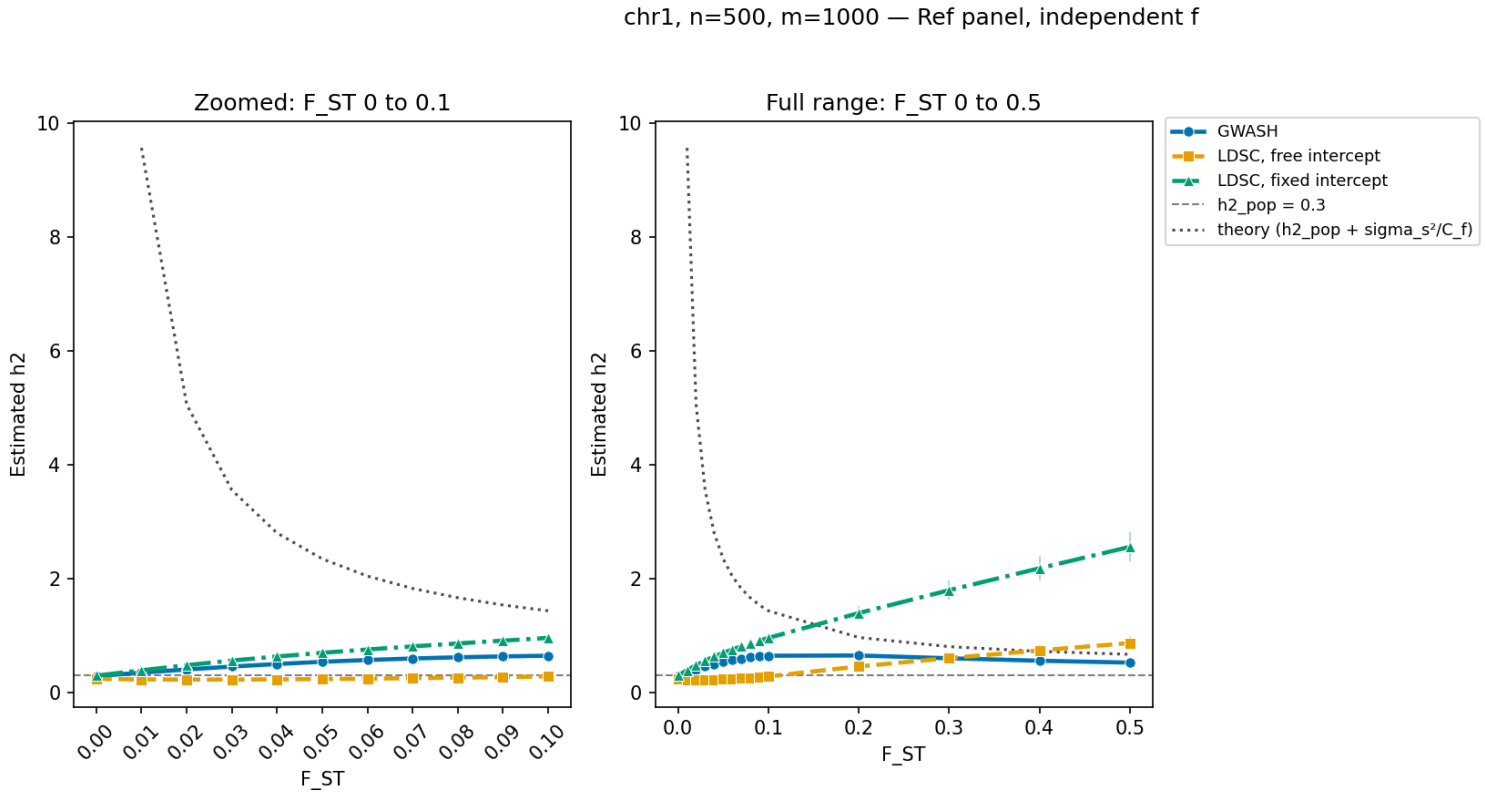
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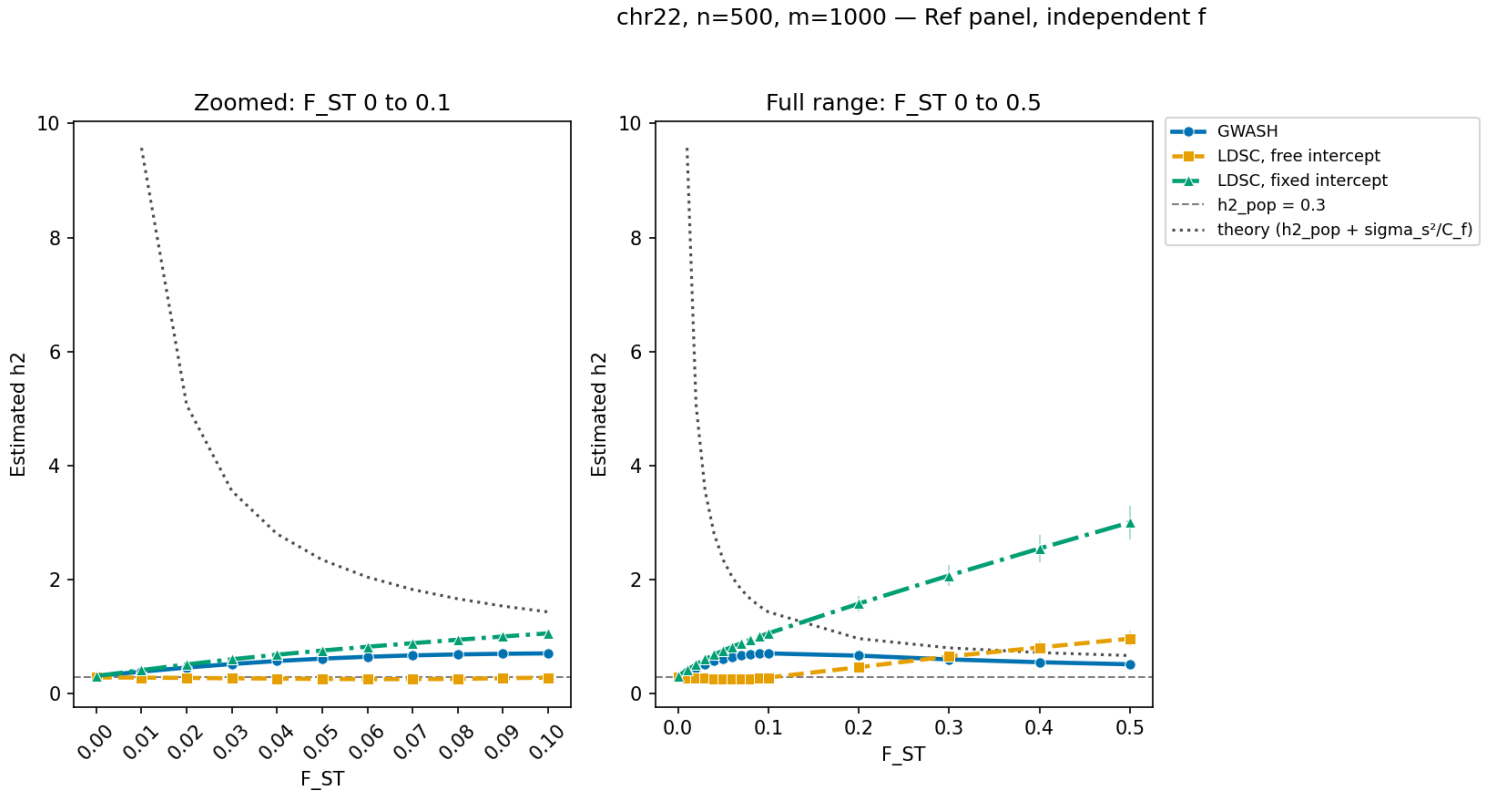
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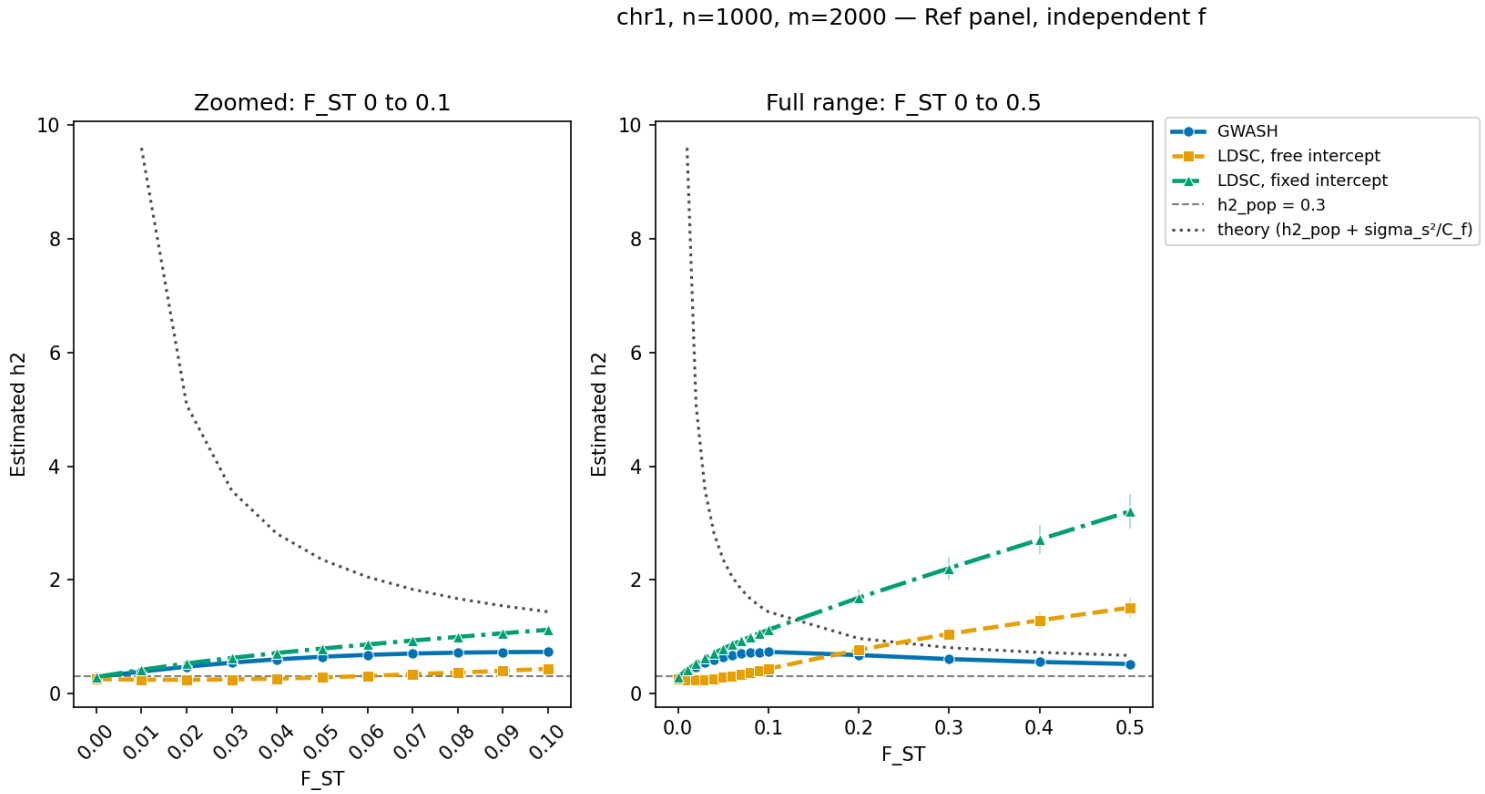
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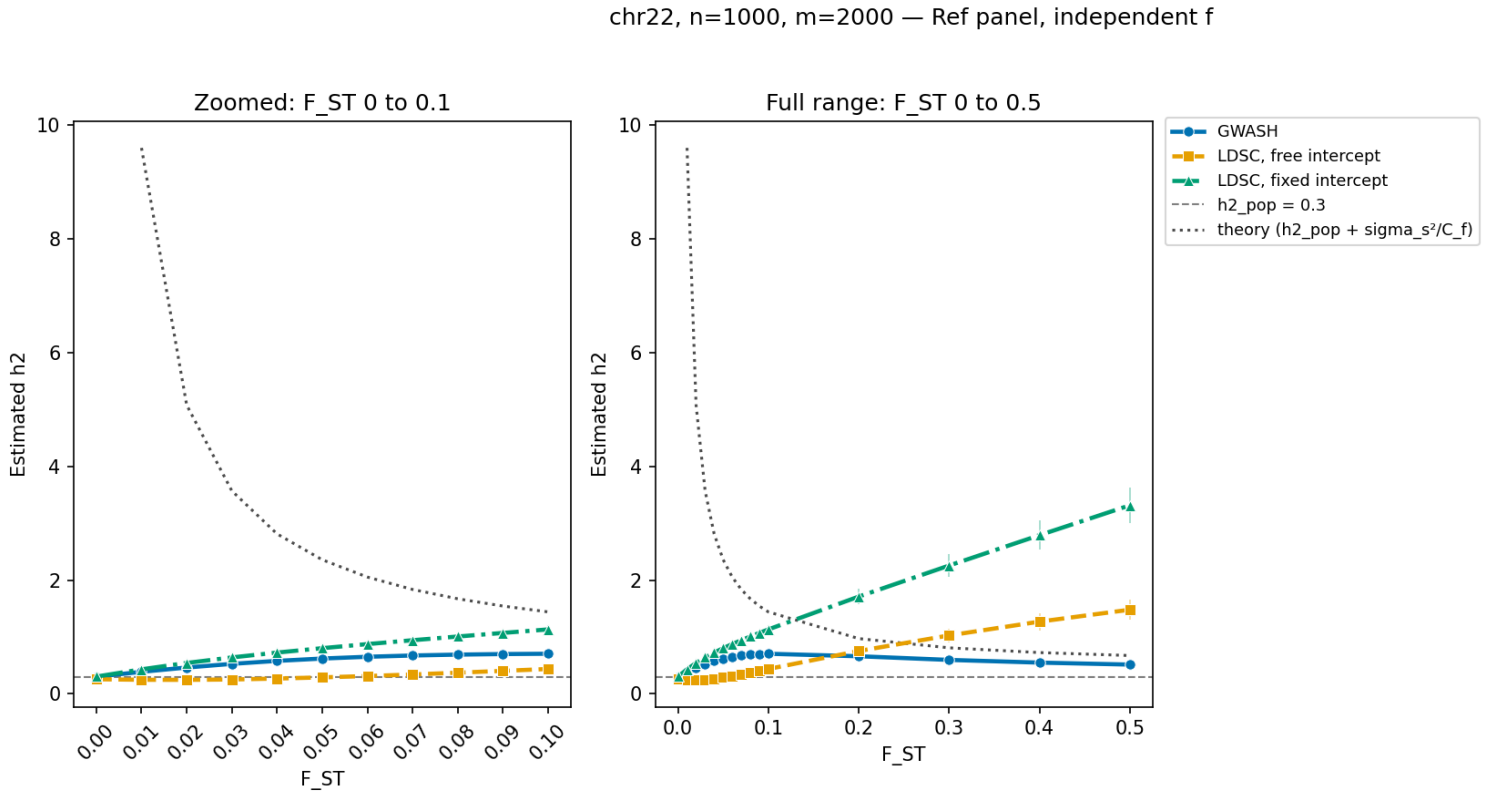
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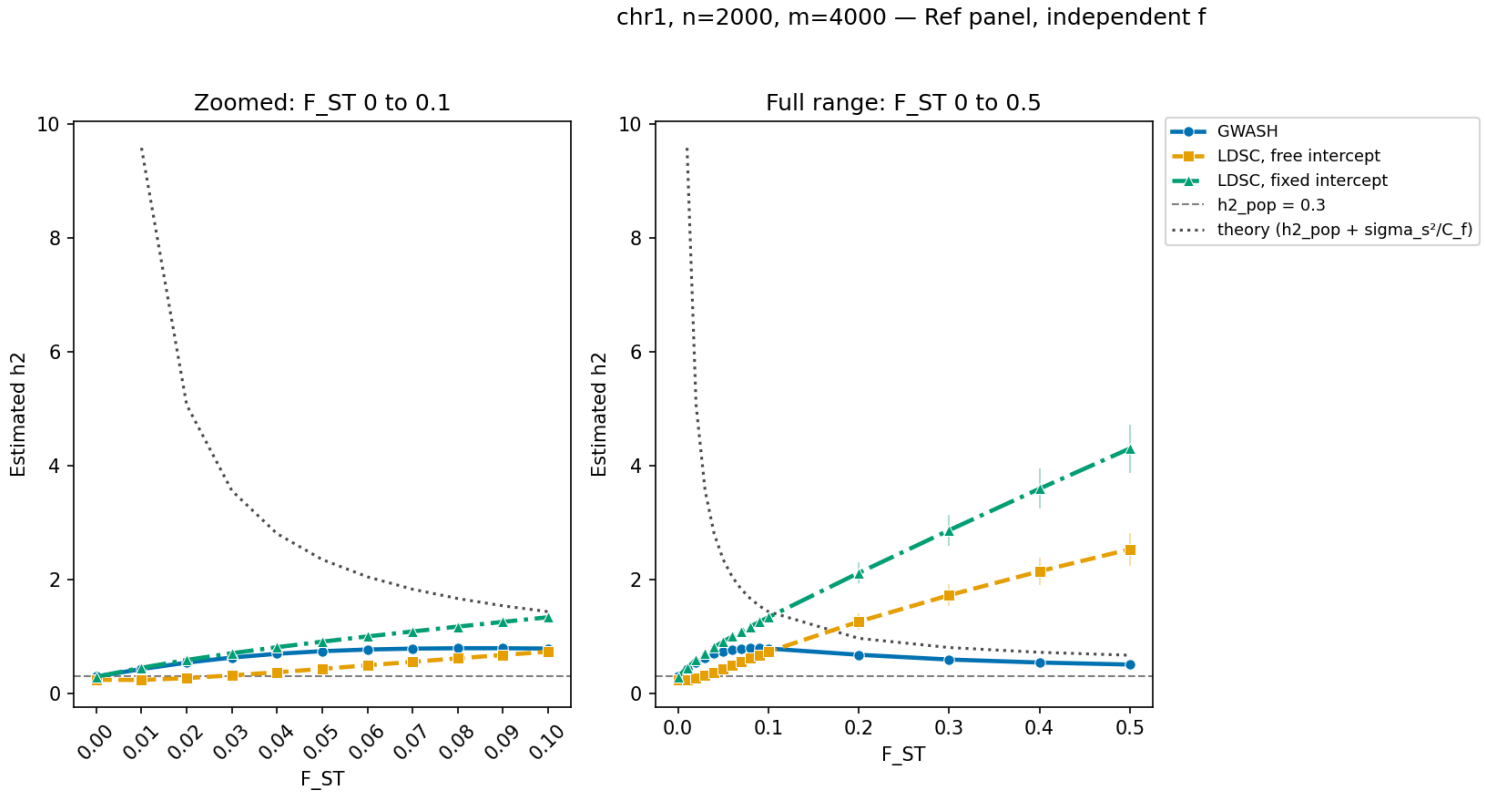
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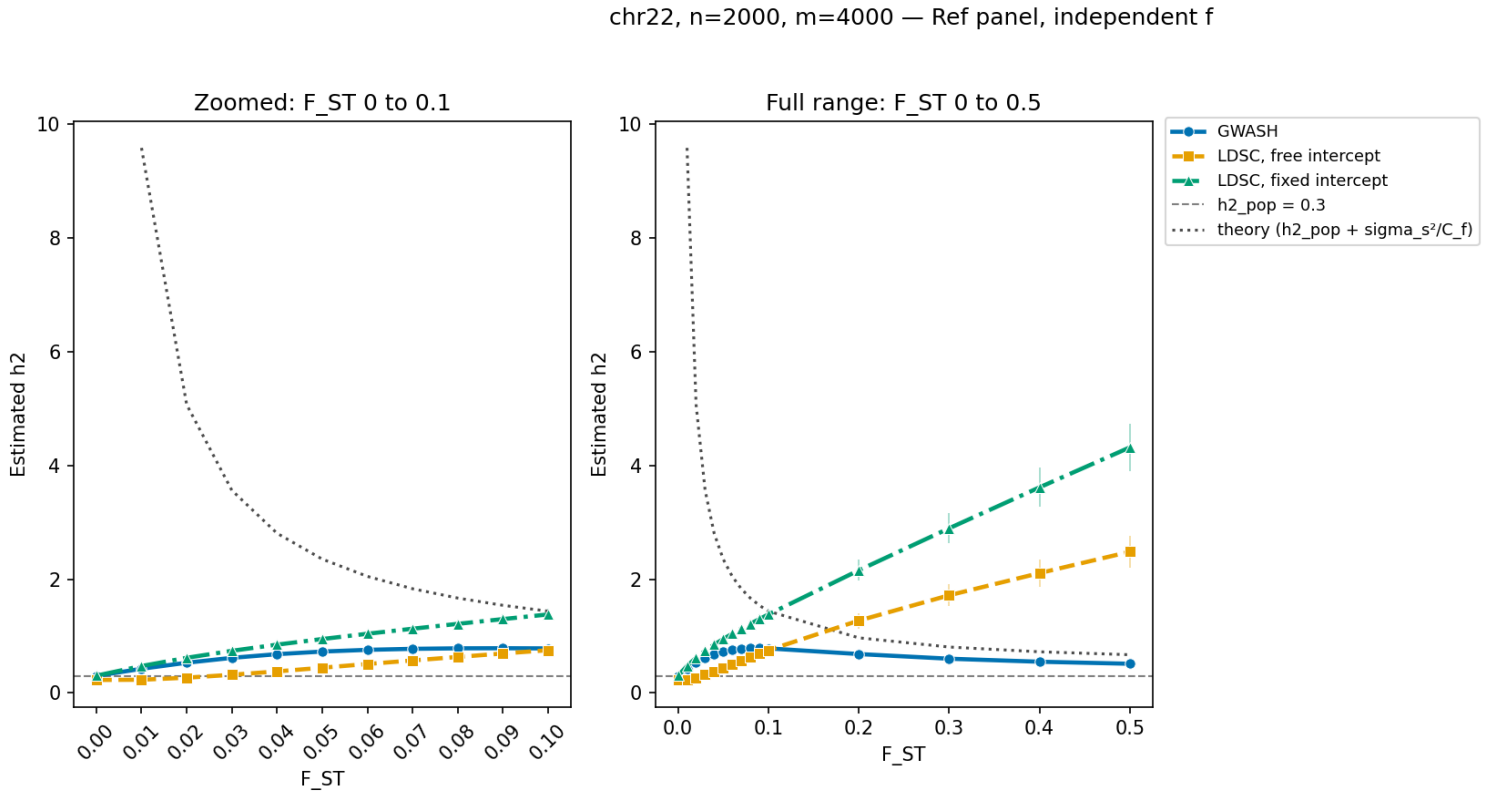
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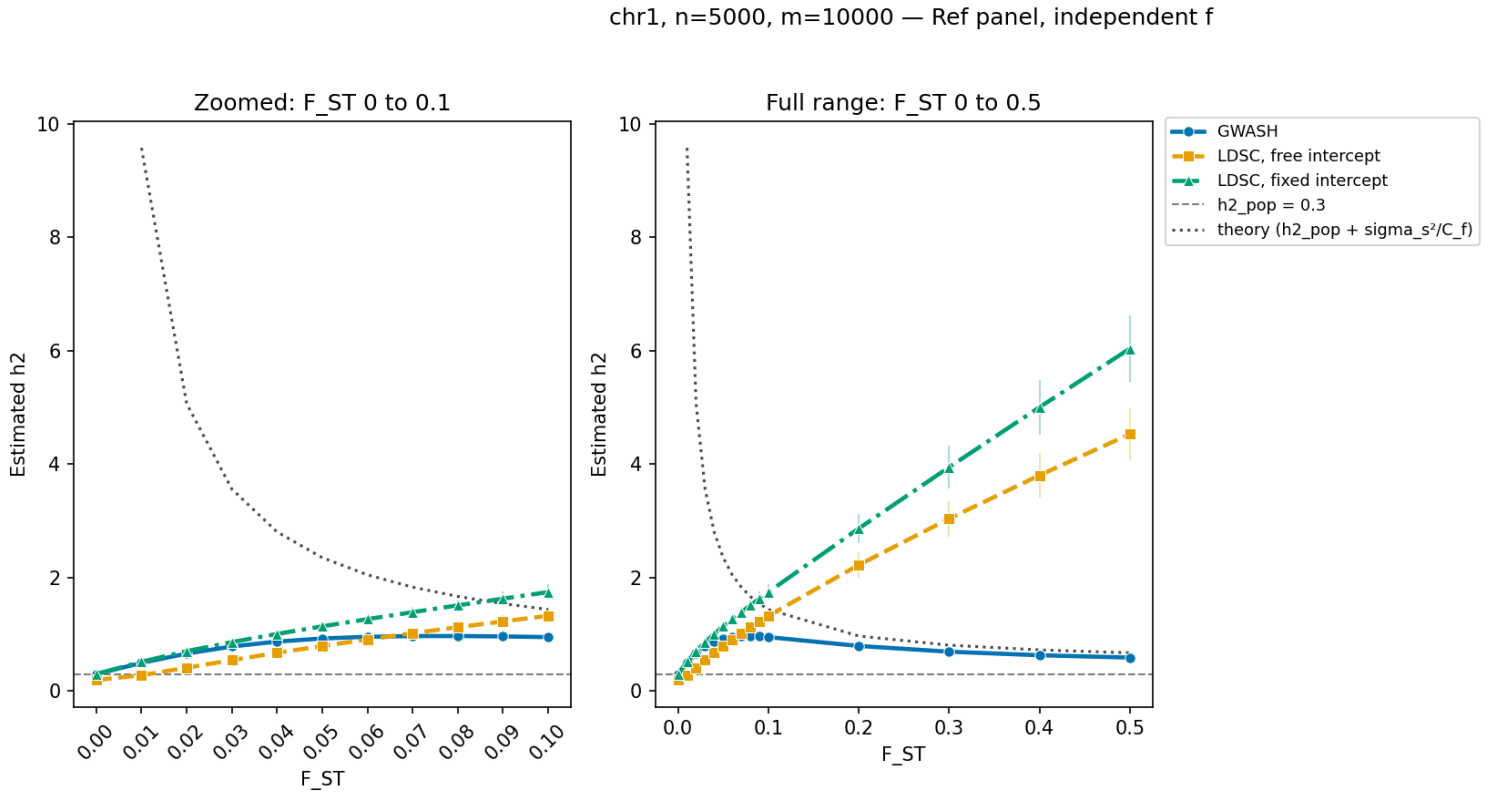
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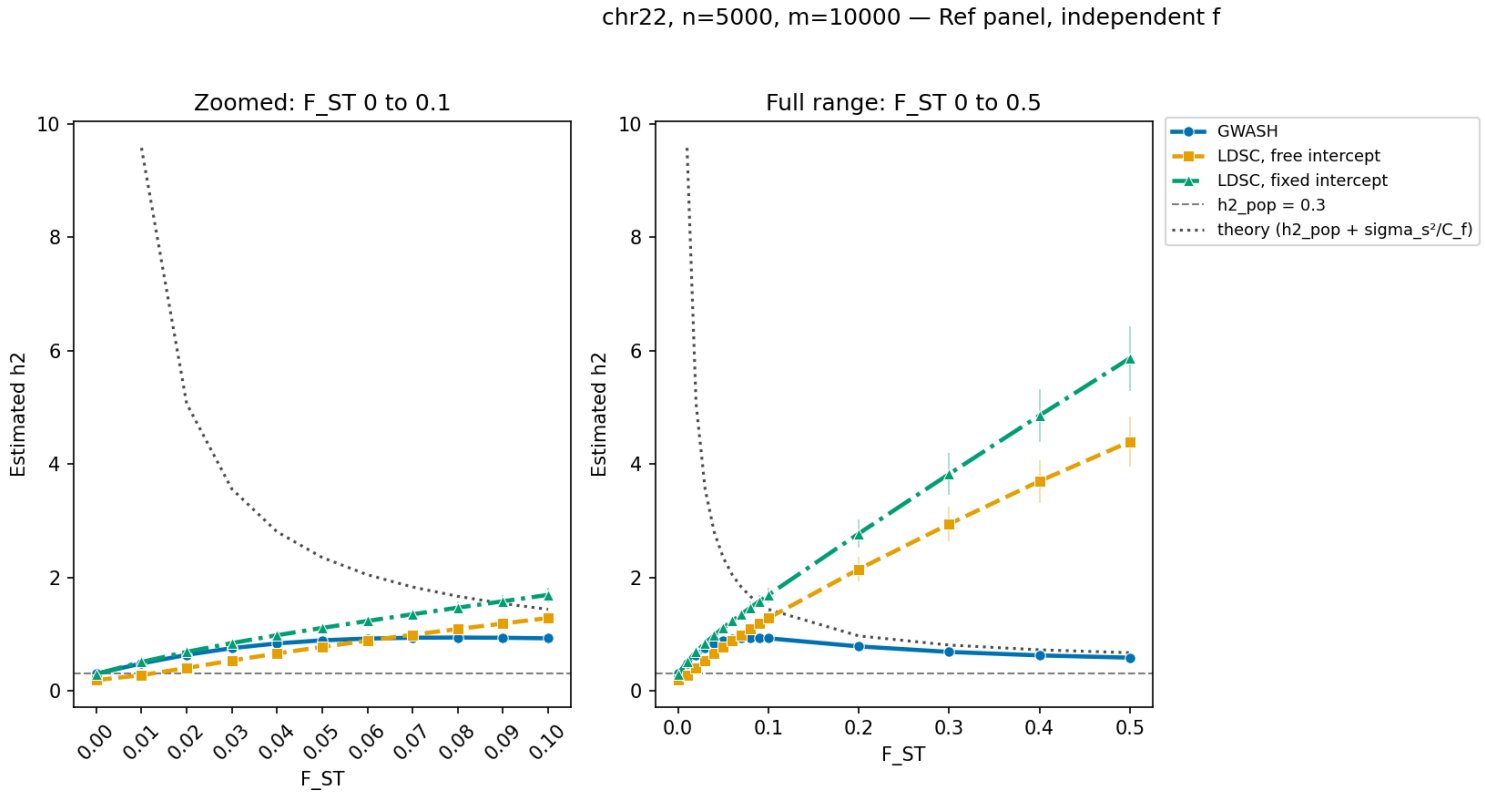
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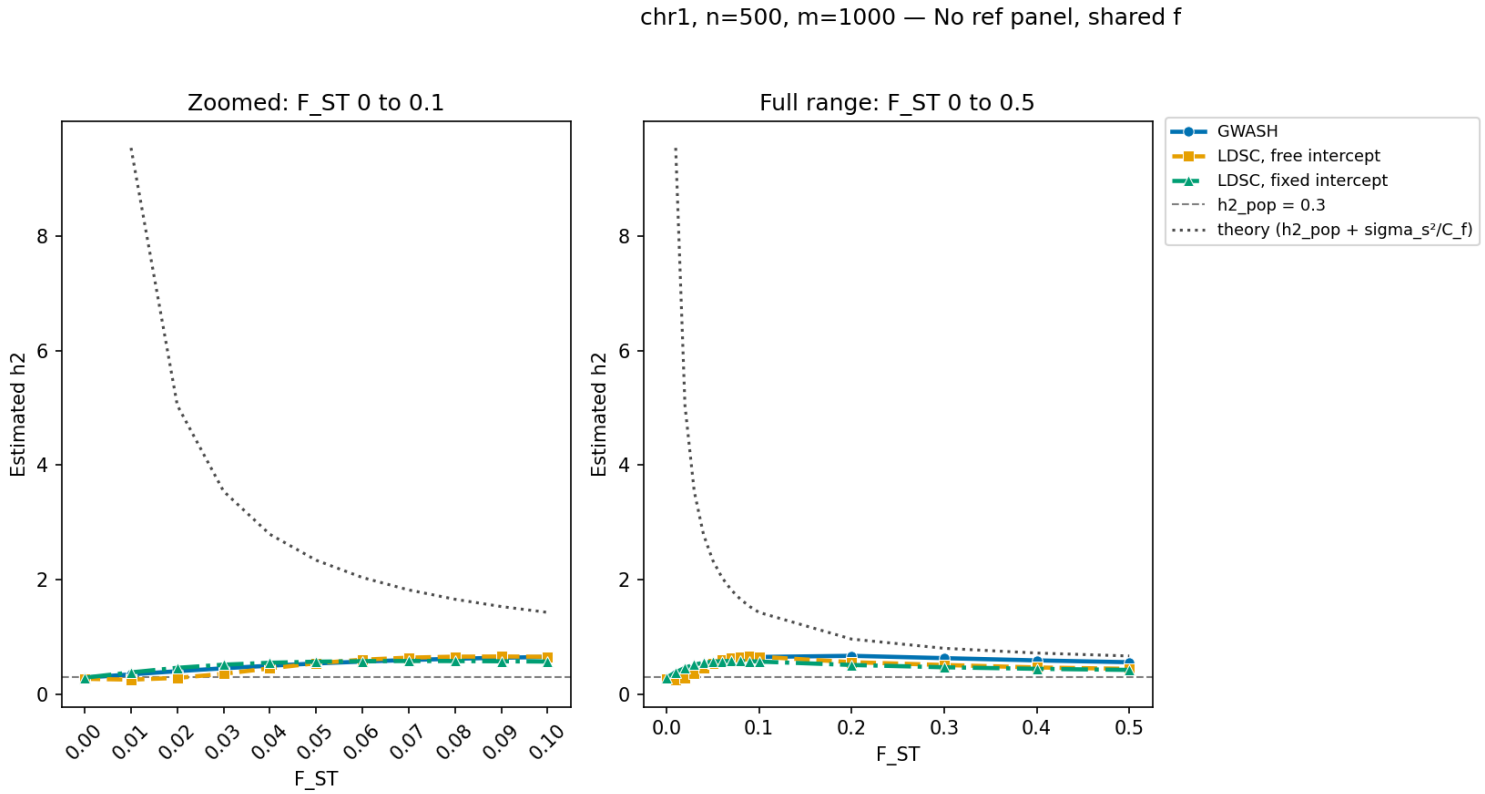
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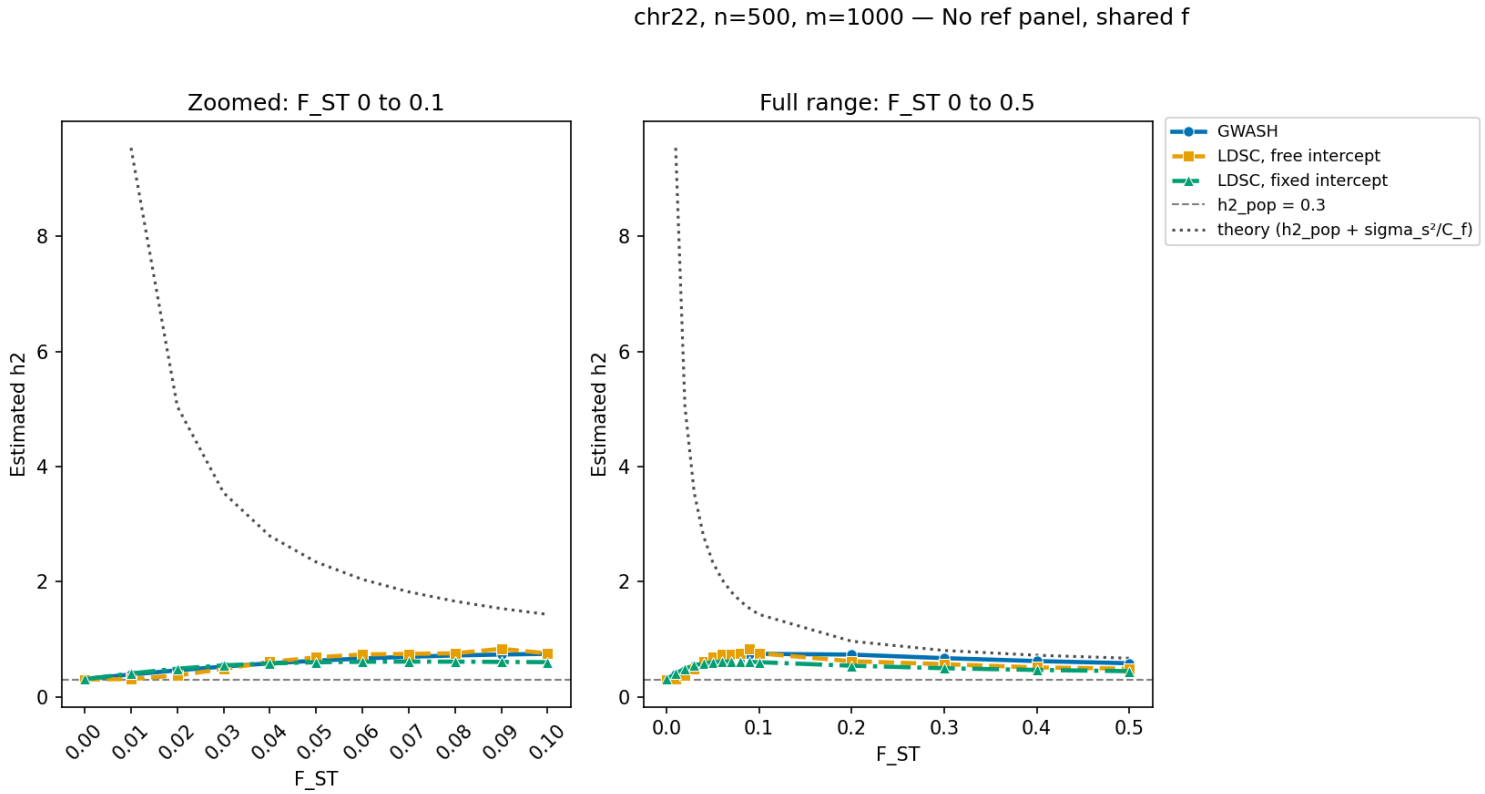
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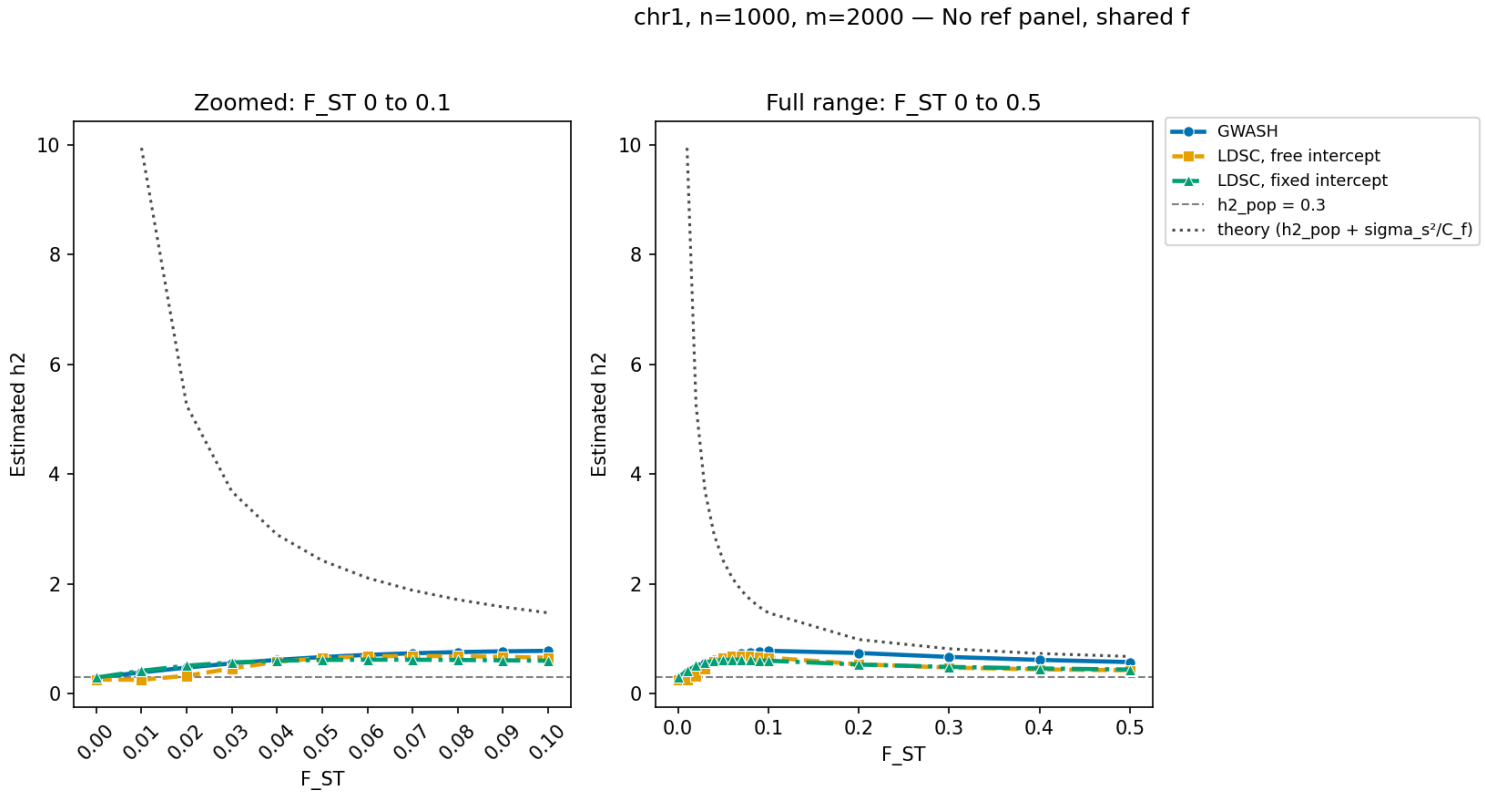
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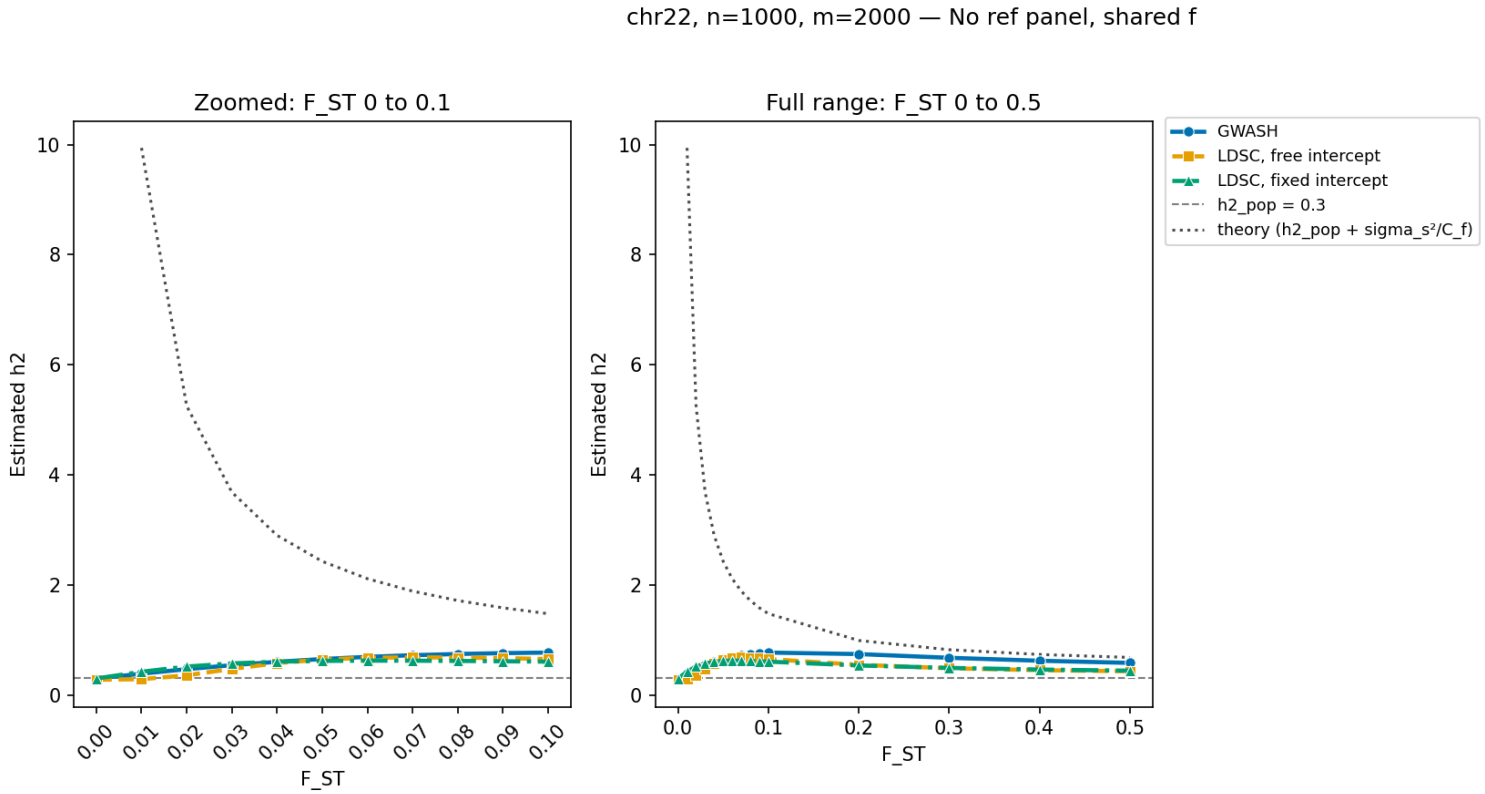
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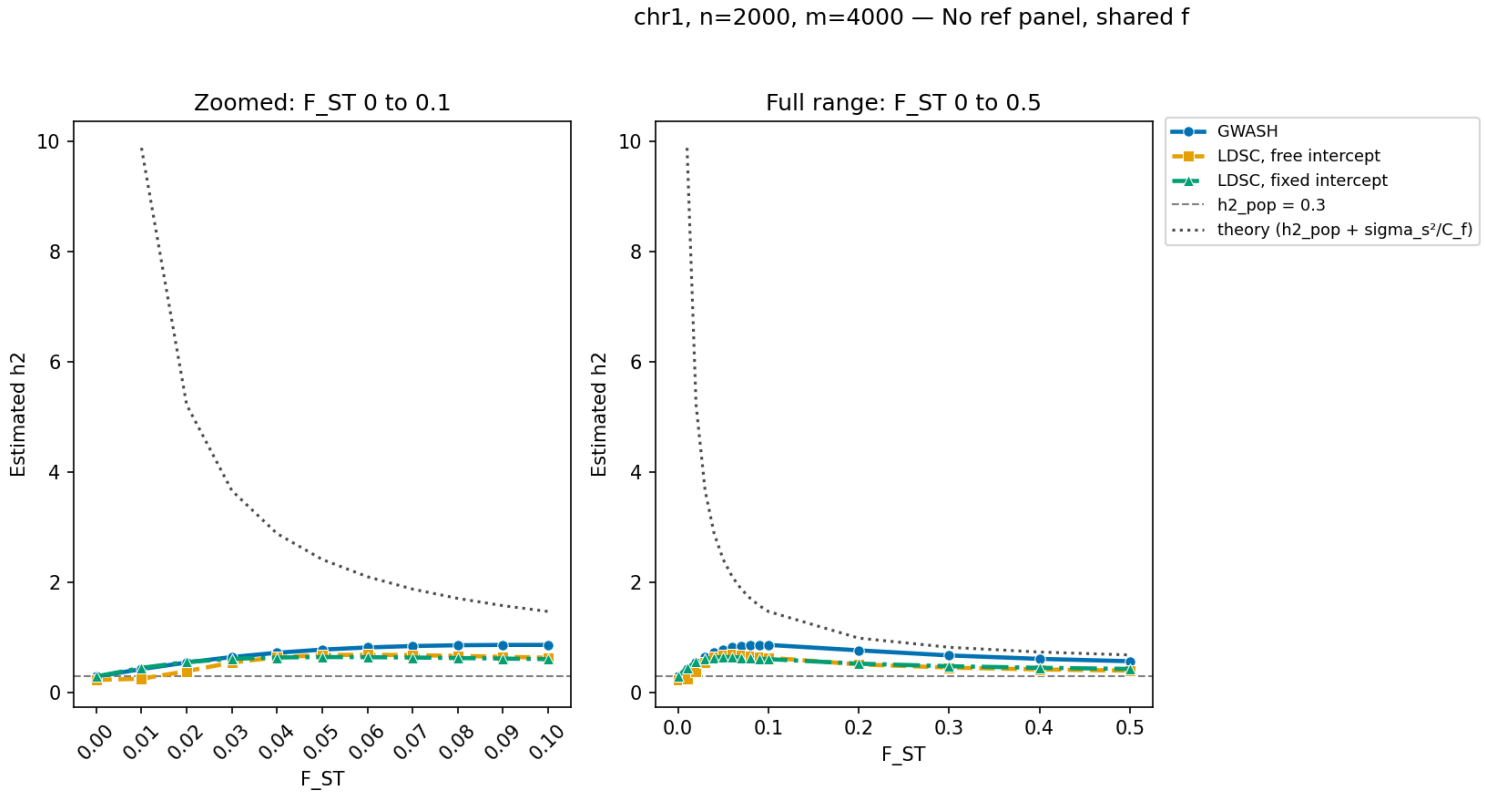
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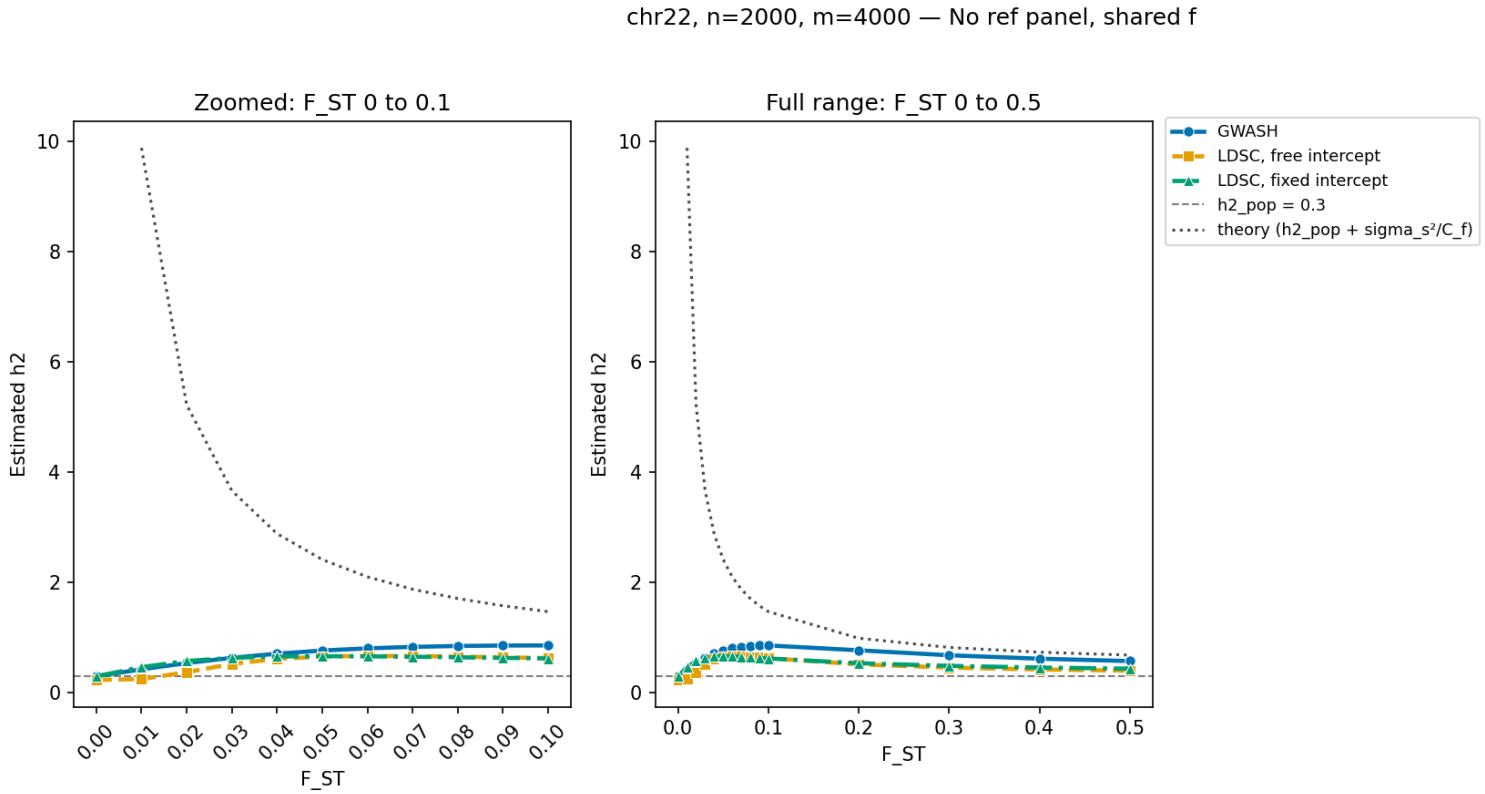
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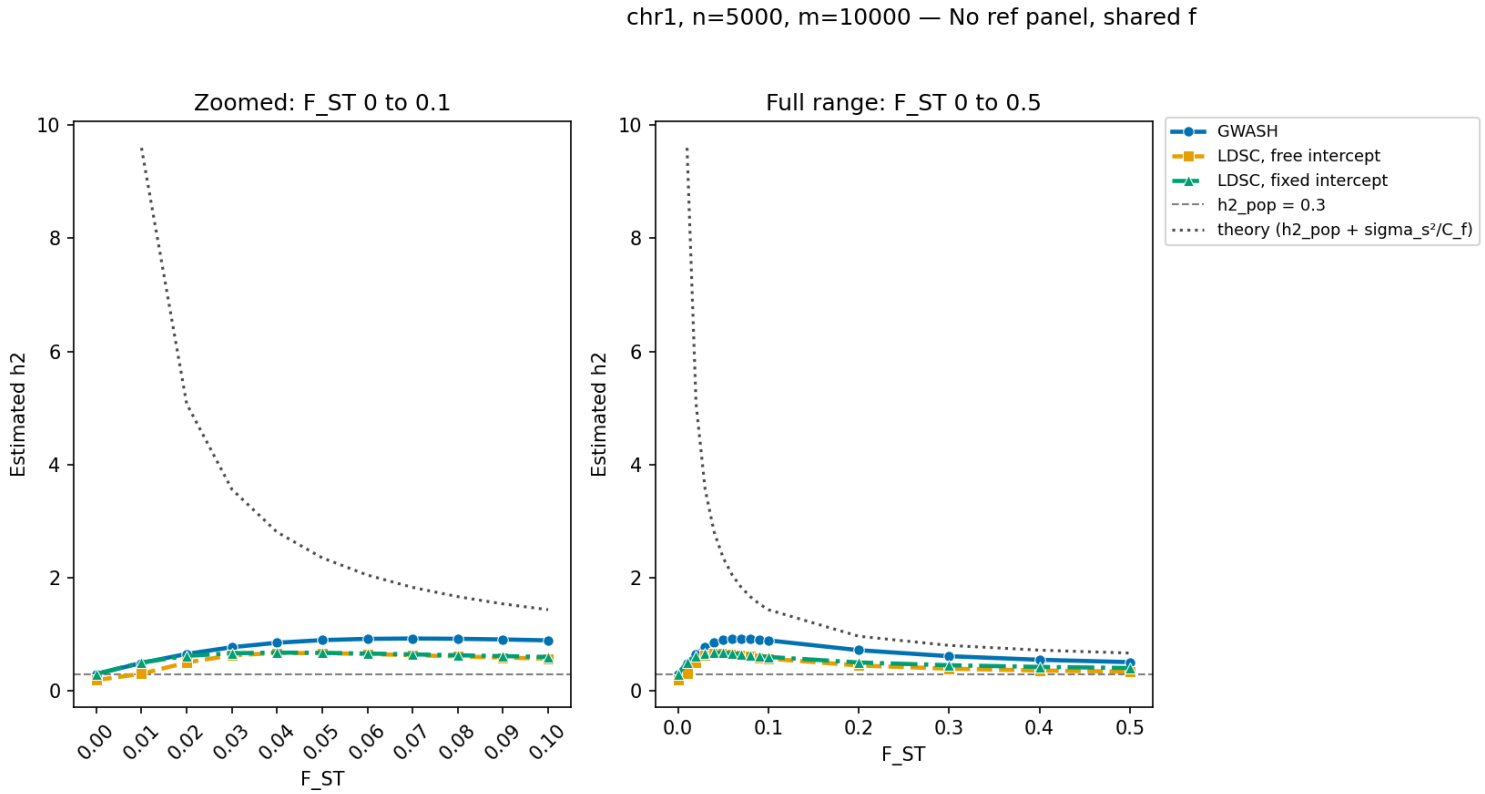
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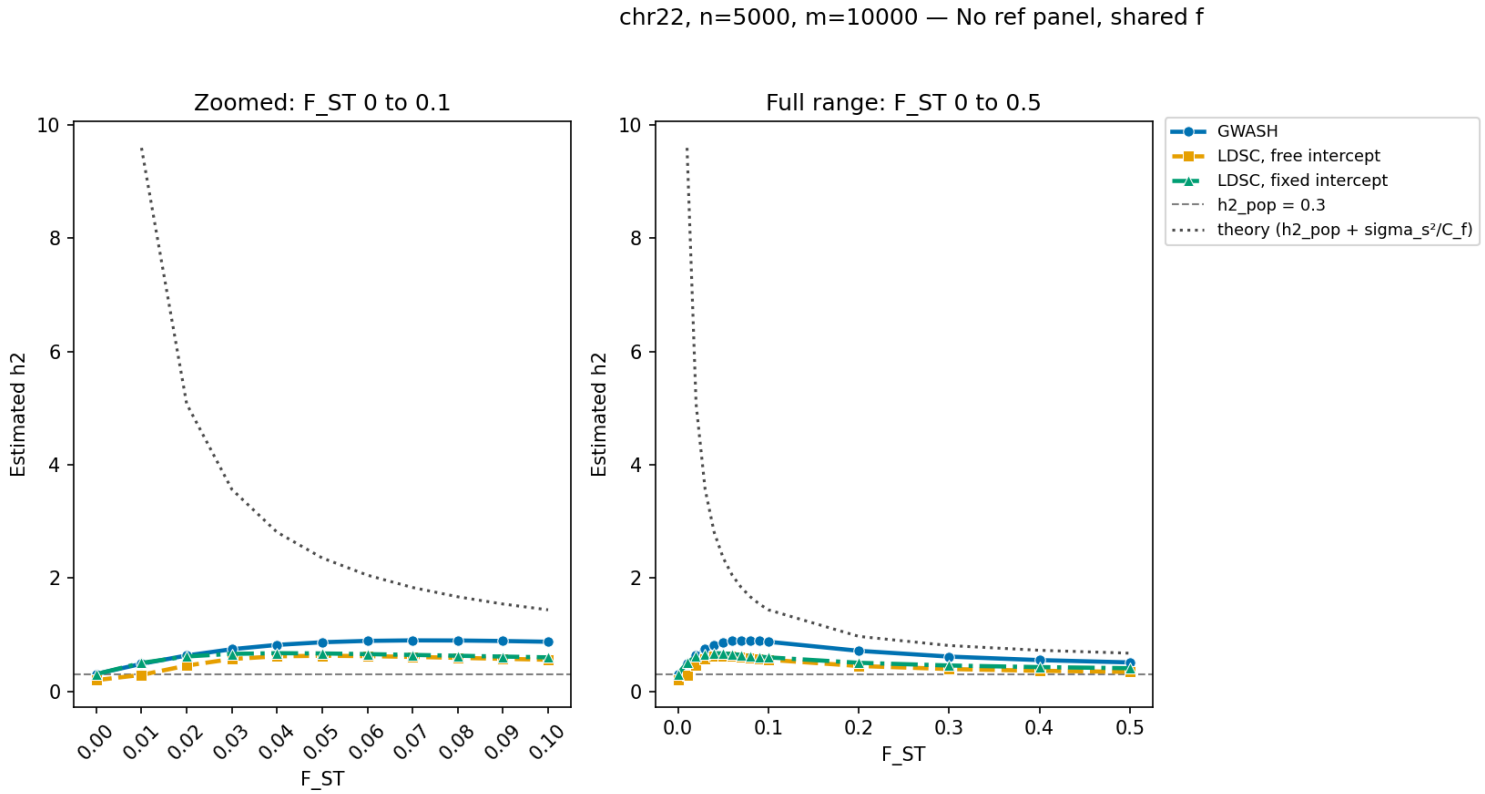
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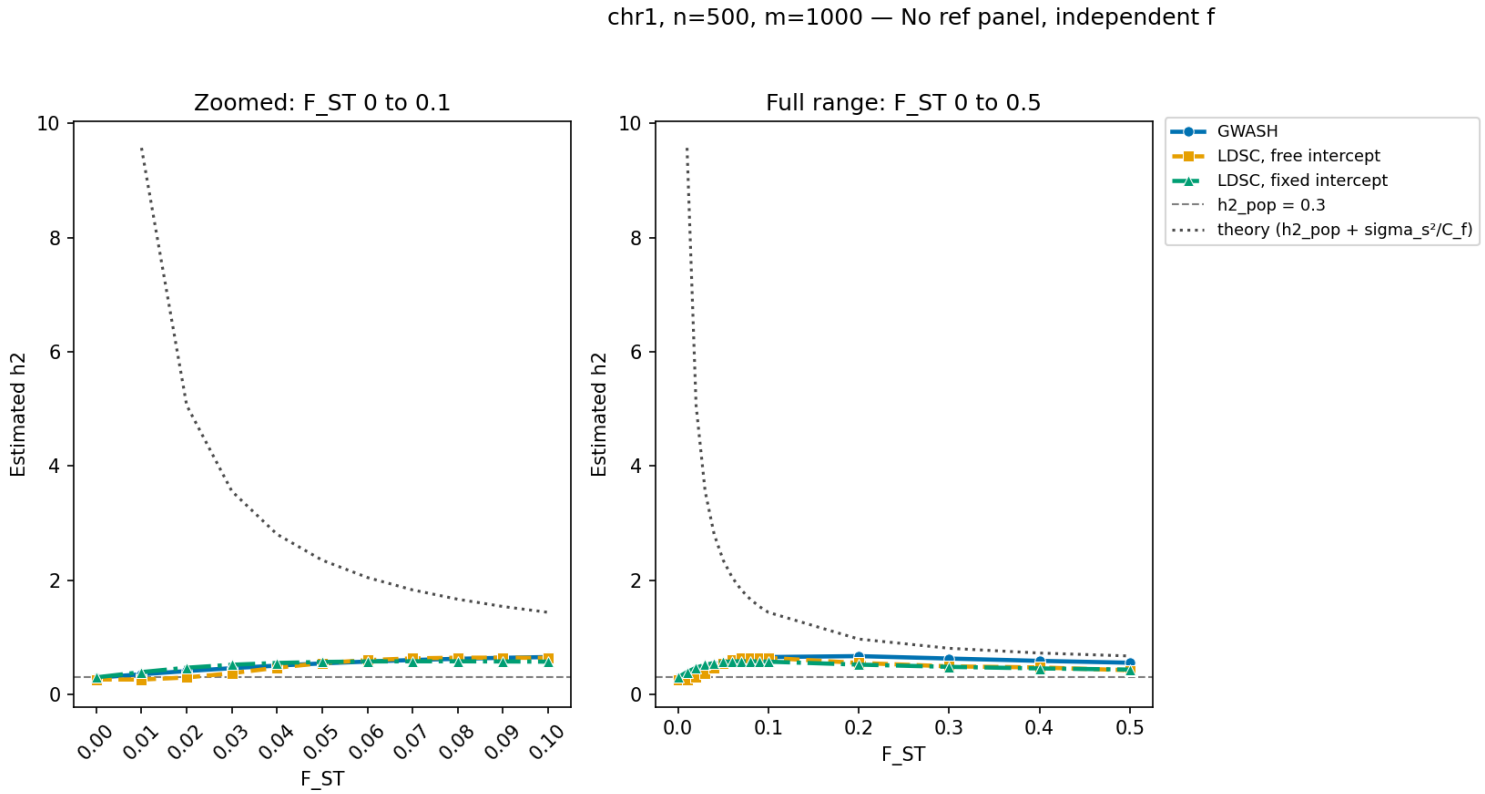
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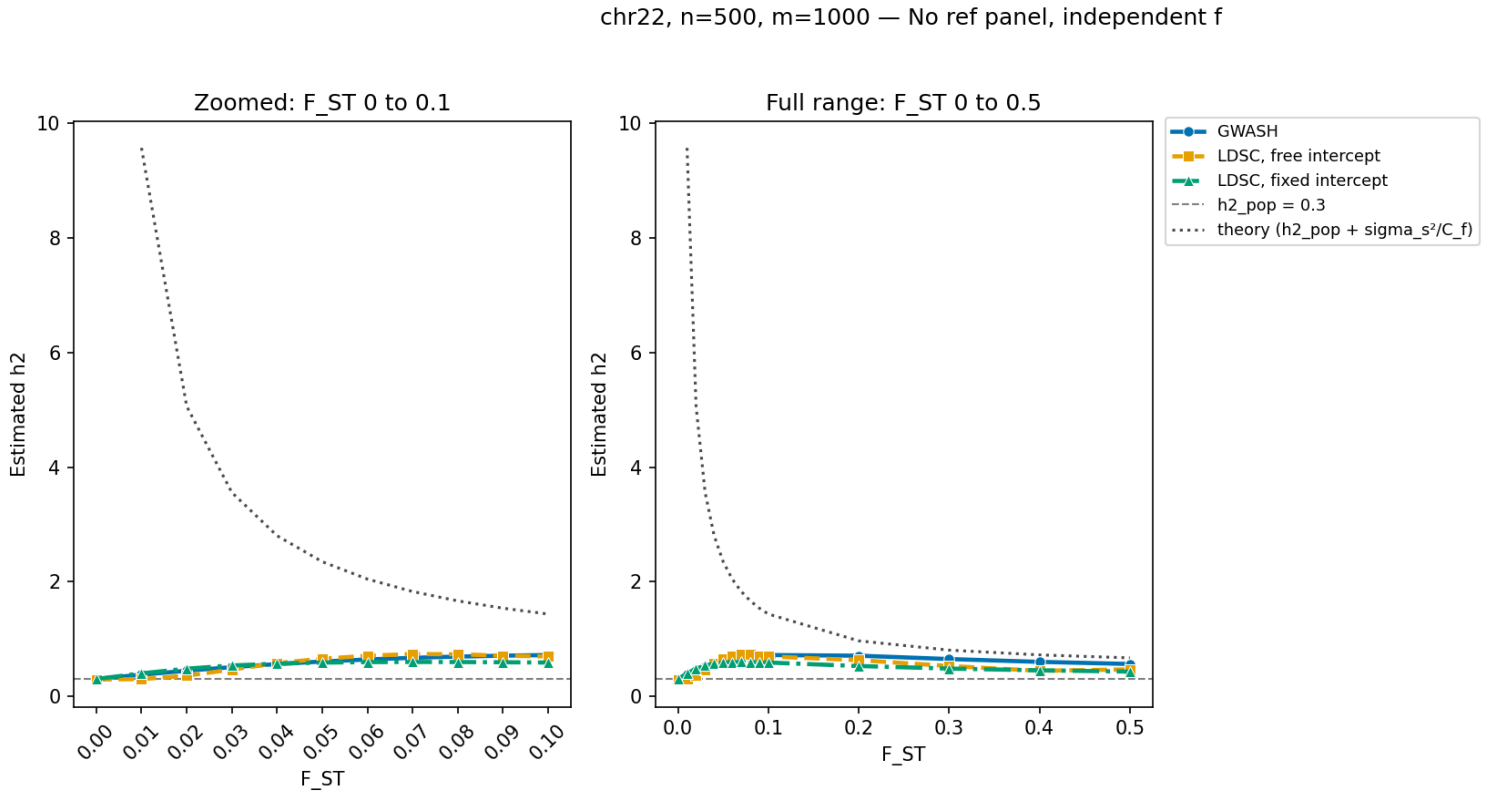
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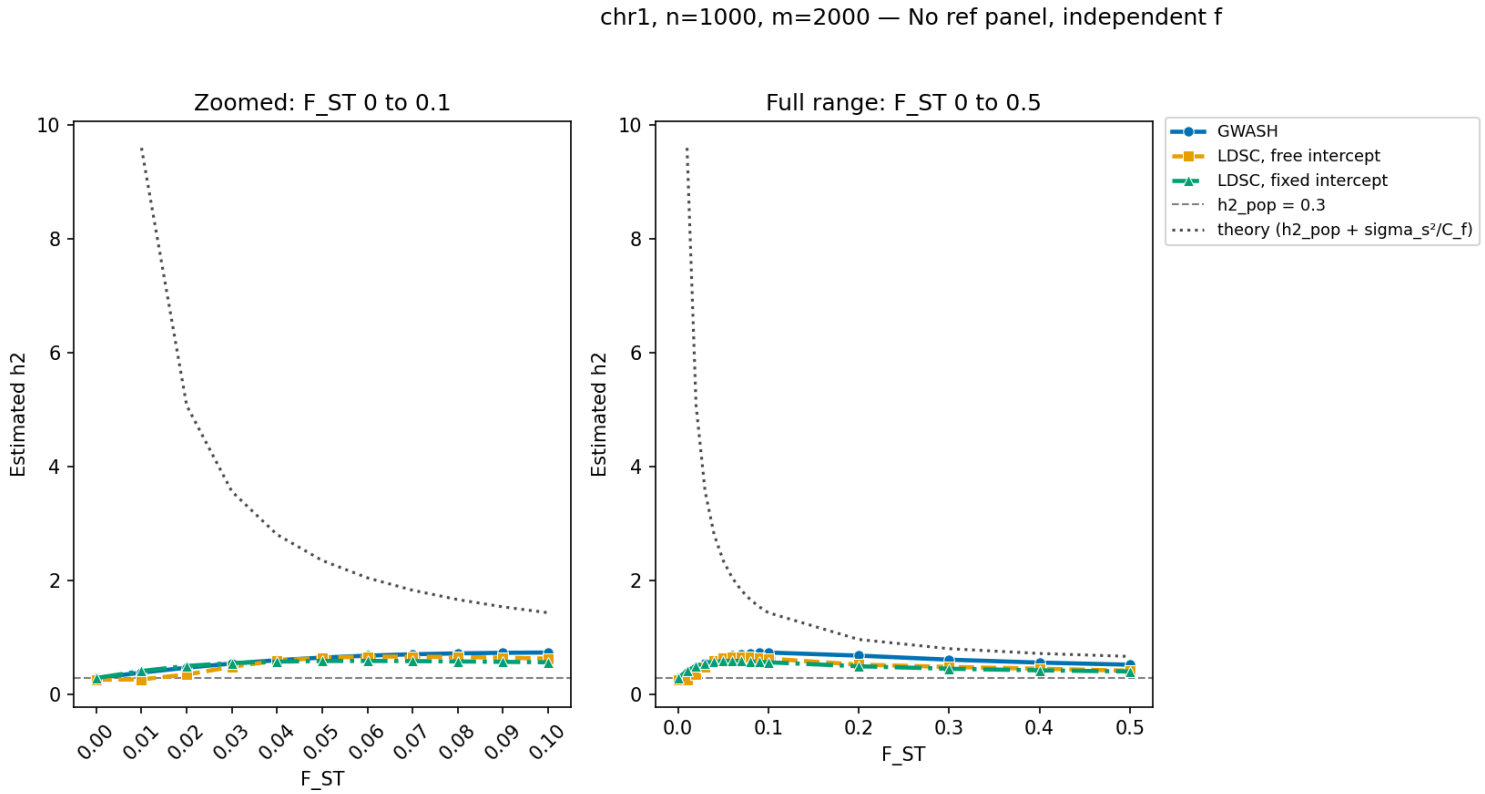
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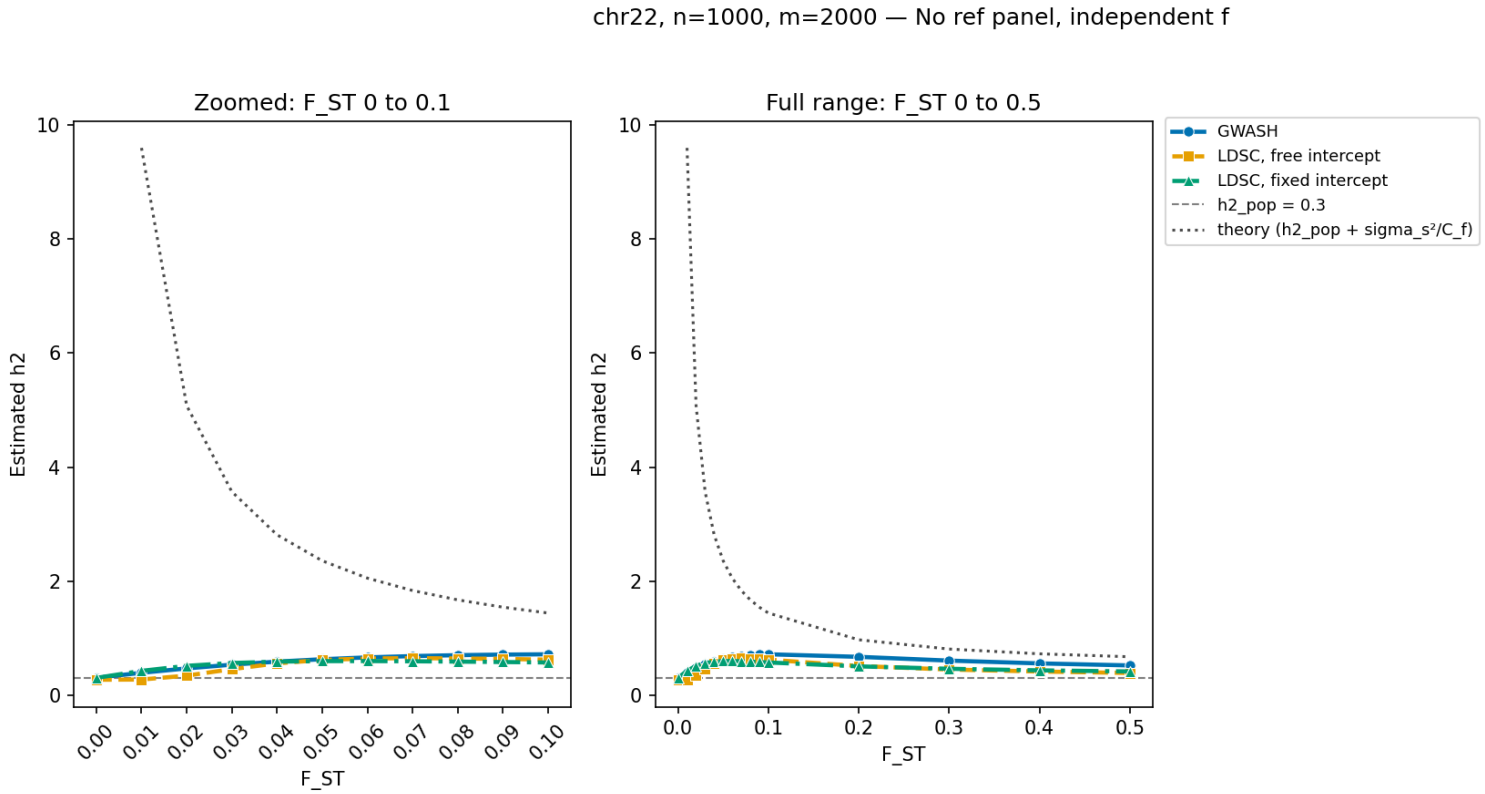
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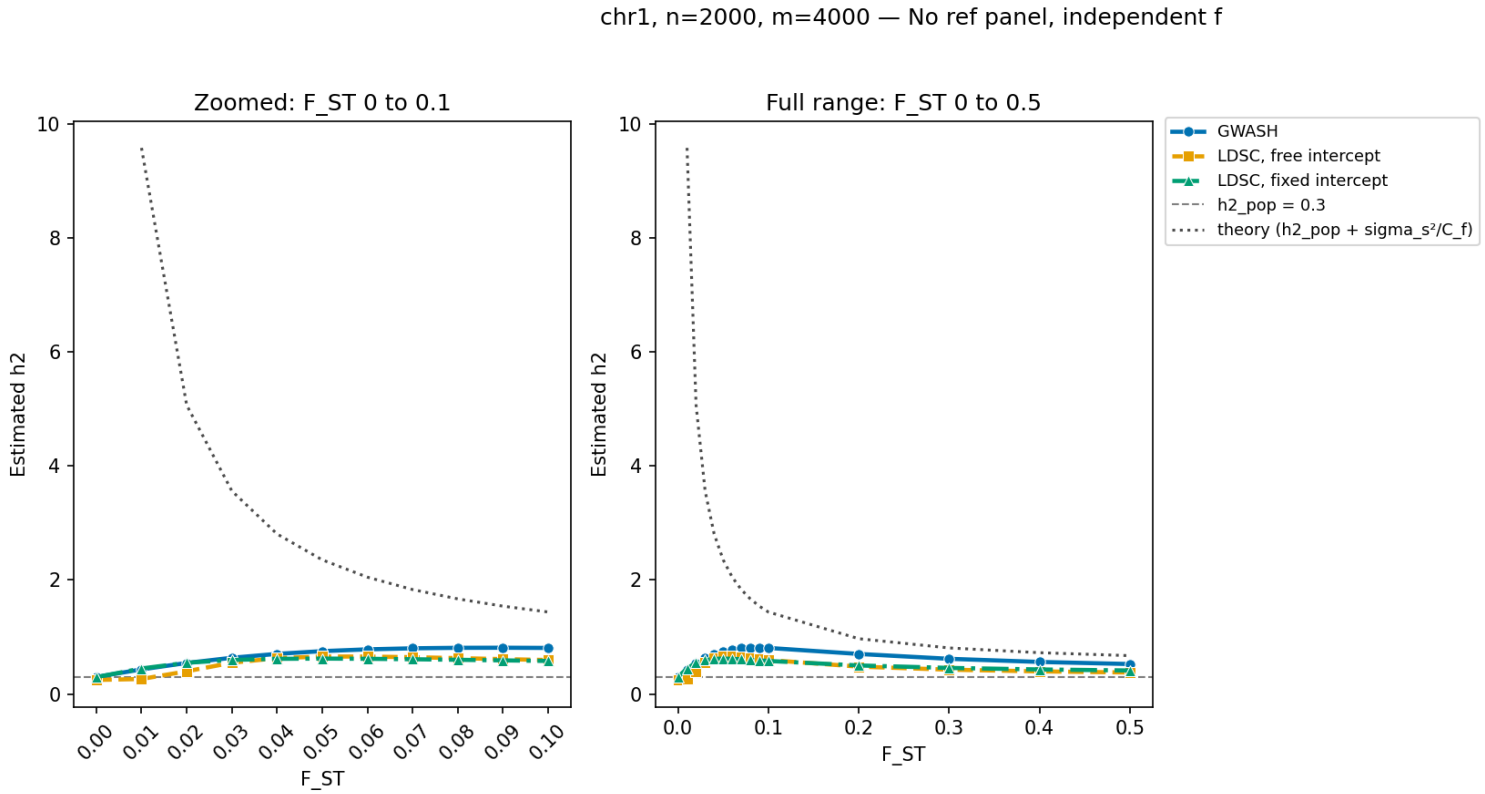
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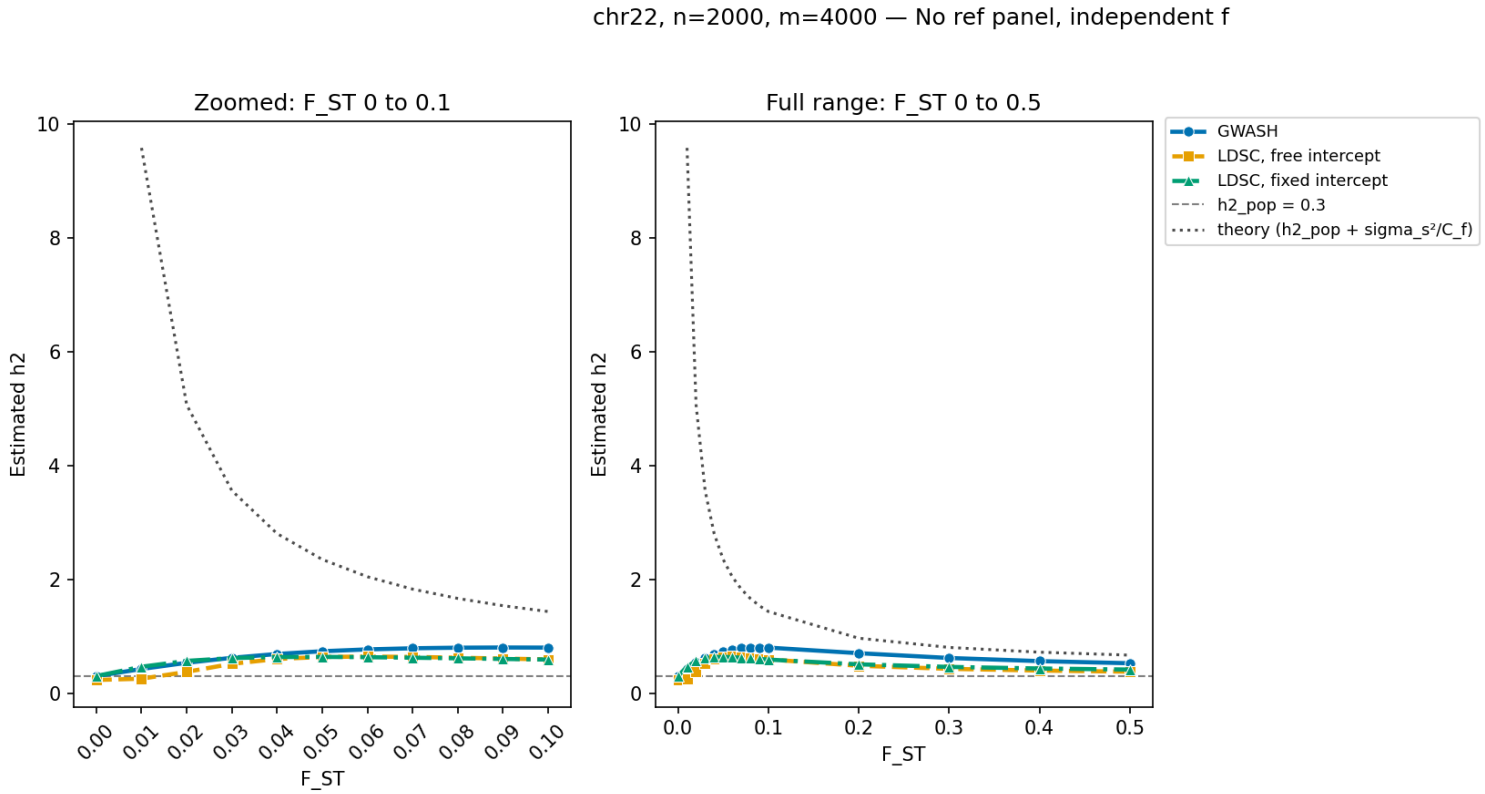
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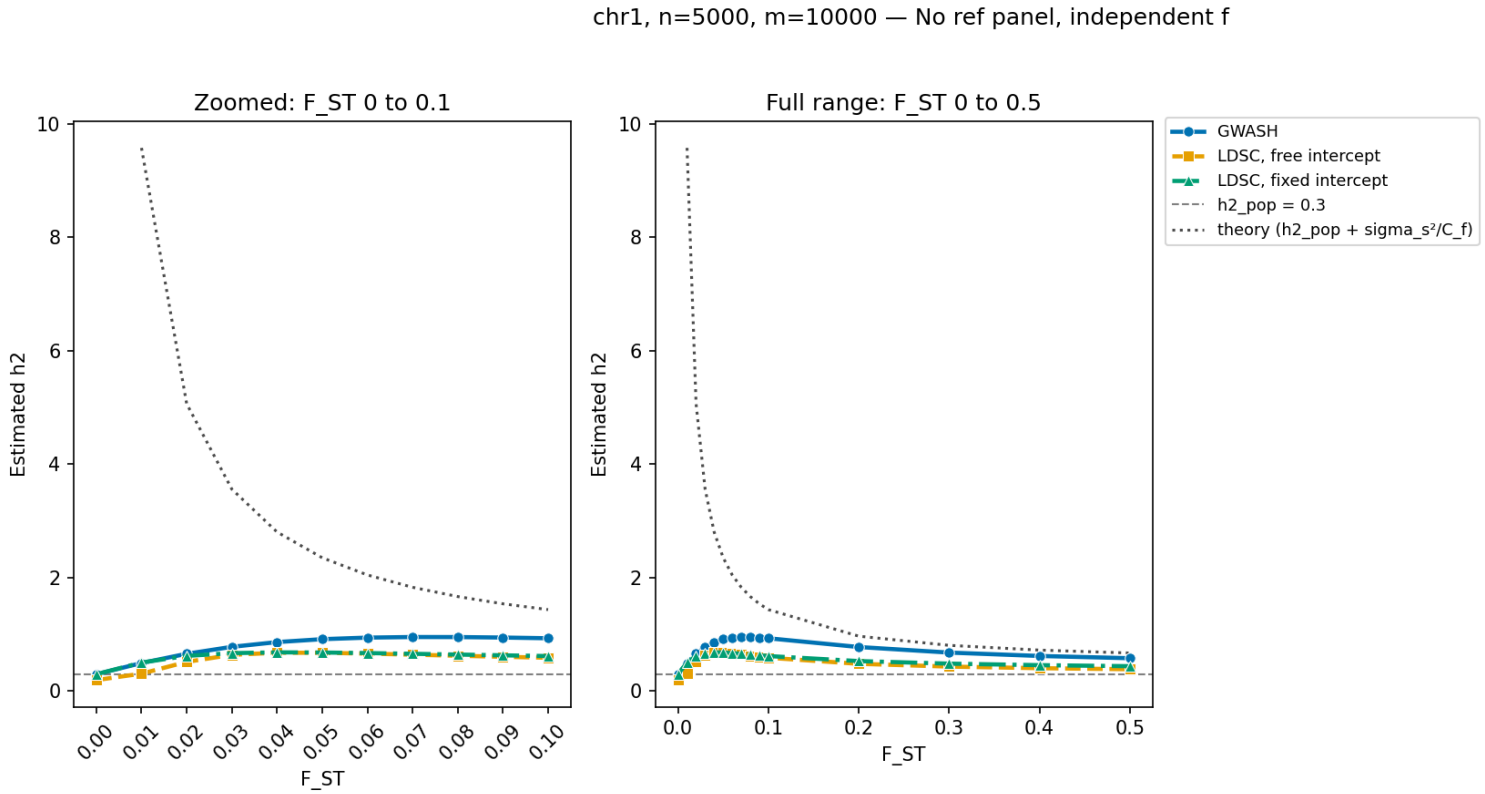
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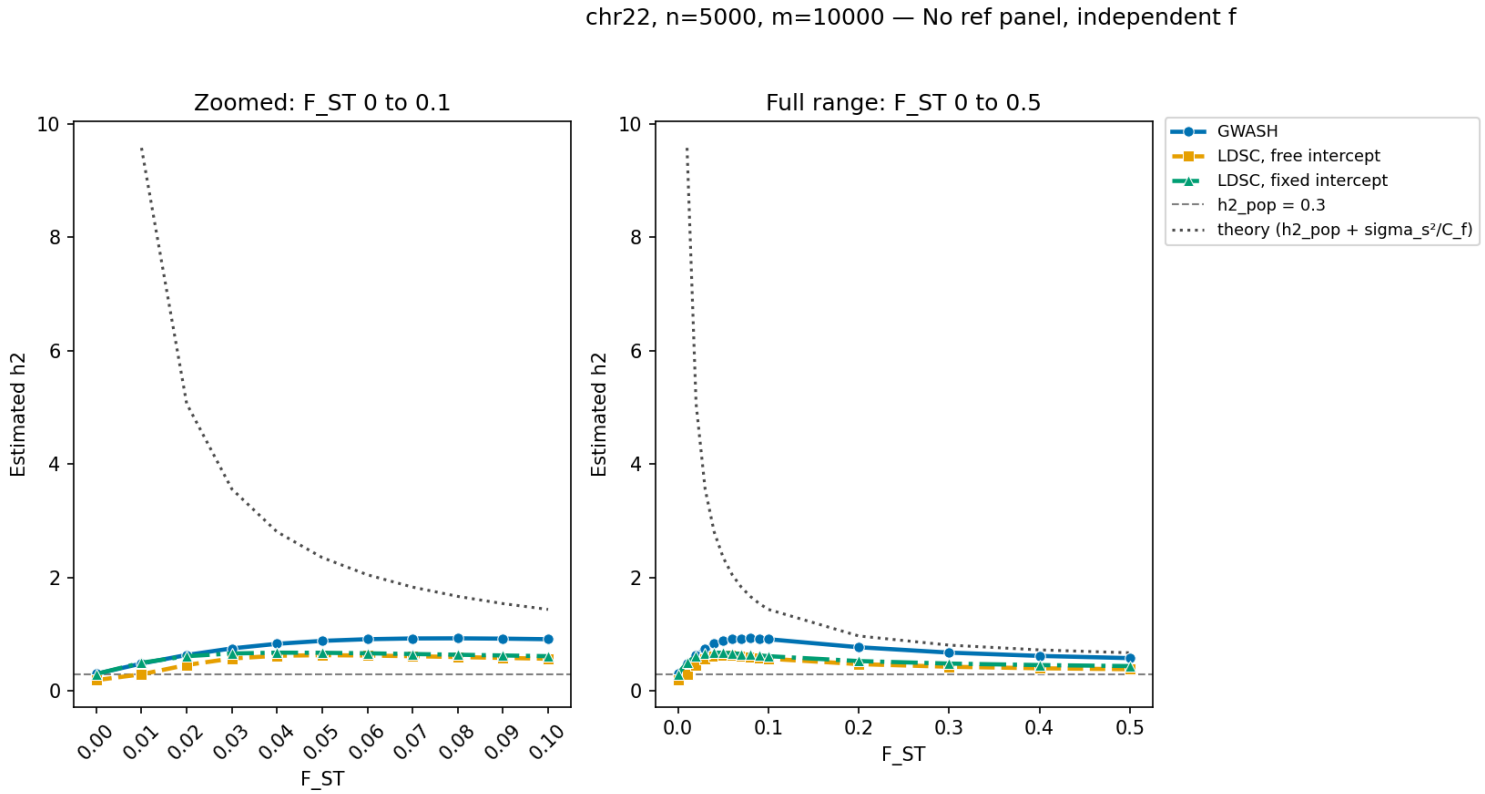
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Discussion

GWASH and LDSC agree when the reference panel is well-specified: with no reference panel at all, or with a reference panel whose f matches the main sample, GWASH and both LDSC intercept choices move together with F_{ST} and track the theory line - see the 'No ref panel' and 'Ref panel, shared f' figures below.

They diverge sharply once the reference panel is mismatched: in the 'Ref panel, independent f' scenario, LDSC's estimates (both intercepts) grow with sample size - approximately 5-7× larger at n=5000 than at n=500 at F_{ST}=0.5. GWASH's estimate in the same scenario stays essentially flat across that same range of scales.

Fixed vs. free intercept matters less than the reference-panel mismatch itself: LDSC's fixed-intercept estimate is consistently larger than its free-intercept estimate in the mismatched scenario (2.56 vs 0.87 at n=500, rising to 6.04 vs 4.53 at n=5000), but both inflate with scale - the intercept choice shifts the magnitude, not the qualitative scale-dependence.

Caveat: GWASH's own bias in the mismatched scenario (roughly 1.7-2× h2_pop, not 1×) means it is not immune to the f-mismatch - it is simply far less sensitive to it than LDSC is, and its error does not compound with sample size the way LDSC's does. This report does not investigate why GWASH's bias plateaus rather than growing with n.